

Asynchronous, Generation-Free ABC: Streaming AMIS Reweighting for Heterogeneous Simulator Workloads on HPC

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Abstract

Sequential approximate Bayesian computation (ABC) advances in generations, and a generation cannot end before its slowest simulation. On a parallel machine this barrier idles every worker that finishes early, and for identically distributed simulation times it costs the straggler factor of the workload: the expected maximum of W times over their mean. We introduce a generation-free ABC sampler. It refits its proposal after every arriving evaluation from the best particles evaluated so far, weights each particle against the mixture of proposals it could have come from, and estimates the posterior afterwards from everything that was simulated. We prove the estimator consistent up to a ratio that measures how faithfully its denominator represents the proposals used, and find it calibrated where a pyABC baseline with the same kernel and schedule under-covers. The barrier’s cost, predicted from the asynchronous run’s timing alone, matches the same sampler run with a barrier over measured costs from $1.2\times$ to $400\times$. On a Cellular Potts tissue simulator the asynchronous sampler completes $2.1\text{--}2.4\times$ the simulations of pyABC in the same hour and ends at a $4.1\times$ tighter tolerance. The advantage turns on at a few milliseconds per simulation. With the starting bandwidth set from the prior-predictive discrepancies, the method recovers a two-parameter tissue posterior tighter than pyABC’s on a larger effective sample. That effective sample size remains limited to a few multiples of the archive size.

1 Introduction

Approximate Bayesian computation (ABC) is the classical route to simulation-based inference (Cranmer et al., 2020): Bayesian inference when a simulator can be run but its likelihood cannot be evaluated (Beaumont, 2019). Its sequential forms, ABC-SMC and ABC-PMC (Sisson et al., 2007; Toni et al., 2009; Beaumont et al., 2009; Del Moral et al., 2012), move a population of particles through a sequence of shrinking tolerances. Every implementation we know of does so in generations: the proposal and the tolerance are updated once the whole population has been simulated. That is a synchronization barrier. On a parallel machine with W workers a generation ends when its slowest simulation does, and every worker that finished earlier waits. If simulation times T_i are drawn from a distribution with mean μ , a generation of W takes about $\mathbb{E}[\max_{i \leq W} T_i]$. The barrier therefore costs a factor $\mathbb{E}[\max_{i \leq W} T_i]/\mu$, the *straggler factor* of the workload, which is 1 for a homogeneous simulator and grows with W and with the spread of the runtime distribution. Simulators worth running on such machines are rarely homogeneous: the cost of a cell-based tissue simulation tracks the number of cells it produces, which varies by orders of magnitude across a prior.

We remove the generation: our sampler refits its proposal mixture after every arriving evaluation, from a sliding archive of the k best particles evaluated so far, and hands the next candidate to whichever worker is free. Because the proposal has moved by the time a result arrives, every particle is weighted against the mixture of proposals actually used, the streaming limit of adaptive multiple importance sampling (AMIS) (Cornuet et al., 2012). Every quantity the sampler uses, including the posterior estimate, is recomputed from the history of evaluations, so there is no shared state to synchronize: workers never wait for one another, a

crashed run is recovered by replaying its log, and the sampler runs unchanged on the island model of the Propulate framework (Taubert et al., 2023).

The practical comparison is with pyABC (Klinger et al., 2018) run with the same kernel and bandwidth schedule. On a tissue simulator our sampler completes $2.4\times$ its simulations in the same hour, ends at a $4.1\times$ tighter tolerance and recovers a tighter posterior on a larger effective sample; under a persistent straggler pyABC’s throughput falls more than a hundredfold and ours by less than half.

Our contributions are the following; C1–C4 label the four claims that §6 tests, one per subsection.

1. A generation-free ABC sampler, smooth-kernel ABC with streaming AMIS reweighting, whose posterior estimate can be replayed from the log of evaluations (§3).
2. A prediction of the barrier’s cost from the asynchronous run’s timing alone, validated against the same sampler run with a barrier over two and a half orders of magnitude of that cost (C1, §6.1).
3. The measurement that the throughput gained converts into a tighter tolerance against pyABC on a tissue simulator, and that the advantage turns on at a few milliseconds per simulation (C2, §6.2; C3, §6.3).
4. Consistency of the estimator up to a ratio that measures how faithfully its denominator represents the proposals used; we decompose that ratio and measure part of it (§4; proofs and a central limit theorem in the appendix). Calibration checks in one and four dimensions and under multimodality, recovery of a two-parameter Cellular Potts posterior, and the two settings that bound the posterior, the starting bandwidth and the archive size (C4, §6.4).

2 Background and Related Work

ABC replaces the likelihood with simulation and a discrepancy: for observed data y , parameter θ and a discrepancy ρ between simulated and observed summaries, the ABC target at bandwidth ϵ is

$$\pi_\epsilon(\theta \mid y) \propto \pi(\theta) L_\epsilon(\theta), \quad L_\epsilon(\theta) = \mathbb{E}_{\rho \sim p(\rho|\theta)}[K_\epsilon(\rho)], \quad (1)$$

with π the prior and K_ϵ the ABC kernel. The classical kernel is the hard indicator $\mathbf{1}[\rho < \epsilon]$, under which a simulation is either accepted or rejected and the output is a population of accepted particles. A smooth kernel such as the Gaussian $\exp(-\rho^2/2\epsilon^2)$ (Wilkinson, 2013; Fearnhead & Prangle, 2012) instead gives every simulation a graded weight and removes the discontinuity in the likelihood approximation. Throughout, ϵ is the bandwidth of this kernel; the *tolerance* a sampler has reached is a separate, empirical quantity defined in §5.3.

Sequential ABC and the generation. Drawing from the prior wastes most simulations, so the sequential family (ABC-SMC, Sisson et al., 2007; Toni et al., 2009; ABC-PMC, Beaumont et al., 2009; adaptive variants, Del Moral et al., 2012; Drovandi & Pettitt, 2011; Lenormand et al., 2013) moves a population of N particles through a sequence of shrinking bandwidths. Generation t fits a mixture proposal q_t to the weighted population of generation $t-1$, draws candidates from it until N have been accepted at the current bandwidth, and weights each accepted particle by $\pi(\theta_i)/q_t(\theta_i)$. The next proposal needs the whole weighted population, so generation $t+1$ cannot begin before the last candidate of generation t has been simulated. On a parallel machine that is the barrier of §1: pyABC (Klinger et al., 2018; Schälte et al., 2022), ABCpy (Dutta et al., 2021) and ELFI (Lintusaari et al., 2018) parallelize the simulations within a generation and wait at its end.

Reusing every draw: AMIS. Adaptive multiple importance sampling (AMIS) (Cornuet et al., 2012) removes the dependence of each weight on one proposal. After T stages with proposals $q_1, \dots, q_T$, every past particle is reweighted against the mixture of all of them,

$$w_i = \frac{\pi(\theta_i)}{\frac{1}{T} \sum_{t=1}^T q_t(\theta_i)} \quad \text{instead of} \quad w_i = \frac{\pi(\theta_i)}{q_{t(i)}(\theta_i)}, \quad (2)$$

the balance heuristic of [Veach & Guibas \(1995\)](#). A particle’s weight then no longer depends on which stage happened to propose it, and particles from every stage contribute to the estimate. AMIS belongs to the adaptive-importance-sampling line of population Monte Carlo ([Cappé et al., 2004](#)), surveyed by [Bugallo et al. \(2017\)](#), and is consistent under regularity conditions ([Marin et al., 2019](#)). Our method takes its streaming limit: a stage is one draw, the proposal is refitted after every arrival, and the mixture in the denominator is the sequence of proposals the run used.

Simulators with parameter-dependent cost. The simulators that motivate this work are cell-based and agent-based models of tissues: stochastic lattice or off-lattice models whose likelihood is intractable and which are calibrated against imaging data through summary statistics such as cell counts and tissue shape, typically by ABC ([Kursawe et al., 2018](#); [Lambert et al., 2018](#)). The Cellular Potts model ([Graner & Glazier, 1992](#); [Hirashima et al., 2017](#)) is the canonical lattice model of this class. Its cost scales with the number of cells the simulation produces, and that number is set by the parameters being inferred, so simulation times vary by orders of magnitude across the prior: the runtime distribution of an ABC run on such a model is wide by construction, which is where a generation barrier costs most (§5.1).

Barrier-free sampling elsewhere. Parallel and off-barrier SMC has been studied for state-space models: the island particle model ([Vergé et al., 2015](#)), asynchronous anytime particle methods ([Paige et al., 2014](#)), and parallel resampling ([Murray et al., 2016](#)). None of these targets ABC, and none combines barrier-free execution with AMIS-style reweighting against the mixture of proposals used. Look-ahead scheduling ([Alamoudi et al., 2024](#)) starts the next generation early on idle workers and so recovers part of the wait, but keeps the generation. Sequential neural simulation-based inference ([Papamakarios & Murray, 2016](#); [Greenberg et al., 2019](#); [Lueckmann et al., 2021](#)) also proceeds in rounds, each of which simulates a batch from the current proposal and retrains before the next is proposed, so every round waits for its slowest simulation as a generation does; the sampler we propose is specific to ABC. The same wait is the straggler problem of synchronous stochastic gradient descent ([Dean et al., 2012](#); [Chen et al., 2016](#)), met there with backup workers or with asynchronous, staleness-tolerant updates ([Recht et al., 2011](#)). Here the estimator is defined on whatever history has arrived, and the order of arrival matters only through the conditional law of each draw and how faithfully the posterior denominator represents it (§4).

3 Method

3.1 Overview

Figure 1 sets the method beside one generation of ABC-PMC. ABC-PMC proposes a population from one mixture, simulates it, waits for the last simulation, and only then forms the next proposal (row a). Our sampler has no population and no generation (row b). Every evaluation that completes is appended to a growing history (step 1). The k particles with the smallest discrepancies form the *archive*, and the bandwidth of the smooth ABC kernel is tightened as the archive improves (step 2). A Gaussian mixture centered on the archive members, each weighted by its fit at the current bandwidth, is the proposal (steps 3 and 4). One candidate is drawn from it and handed to the worker that has just become free (step 5). By the time that candidate’s result arrives the proposal has changed, so the candidate is weighted against the mixture of proposals it could have come from; this is the balance heuristic of AMIS (§2) applied one draw at a time (step 6). The posterior is not read off a final population: after the run, every evaluated particle is reweighted against the proposals that were in use over the whole run.

Every quantity the sampler uses is recomputed from the history it is handed. There is no population to keep consistent and no shared mutable state, so workers never wait for one another: each updates from whatever history it has received, a crashed run is recovered by replaying its log, and the posterior estimate is the same whether it is formed during the run or afterwards. The one exception is a small buffer of past proposals that the running sampler keeps for the weight of step 6; it affects which parents are chosen next, not the estimate.

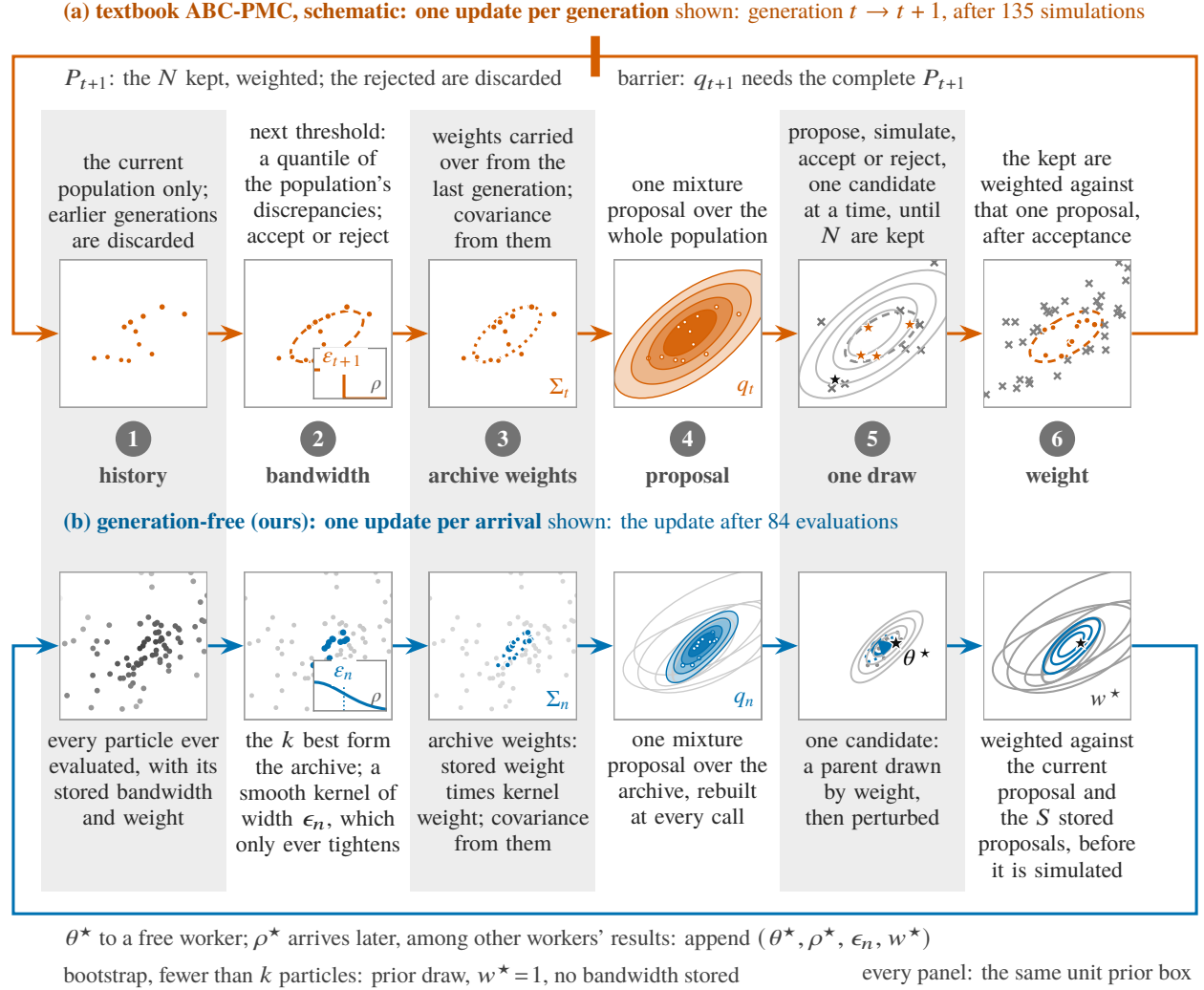


Figure 1: One generation of ABC-PMC (a) beside one update of our sampler (b), step by step on a two-dimensional toy problem. ABC-PMC proposes a whole population from one mixture, simulates all of it, keeps the accepted particles and only then forms the next proposal, so a generation ends with its slowest simulation. Our sampler appends every completed evaluation to a history, rebuilds the archive, the bandwidth and the proposal from that history at every call, draws one candidate and hands it to the free worker; the candidate is weighted against the current and the stored proposals rather than against a single one. Row (a) is a schematic of textbook hard-threshold PMC, not the smooth-kernel pyABC baseline of §5.2. The glyphs are explained in Appendix C.

The *archive weights* $\tilde{W}_j$ shape the next proposal. The *parent weight* w^* stored with each particle is used only inside the running sampler. Only the *posterior weights* $W_{i,n}$, computed afterwards over the whole history, are reported. Sections 3.2 to 3.5 follow the figure's steps; §3.6 assembles them.

3.2 History and archive (steps 1 and 2)

Write the history at call n as

$$\mathcal{H}_n = \{(\theta_i, \rho_i, \epsilon_i, w_i)\}_{i=1}^n, \quad (3)$$

where θ_i is the parameter, ρ_i the discrepancy, ϵ_i the bandwidth in force when θ_i was proposed, and w_i its parent weight (§3.5). The archive A_n is the set of the k history members with the smallest discrepancies.

Until the history holds k members the sampler emits uniform prior draws, with $w_i = 1$ and no bandwidth, so that the first archive can form. The history is append-only, and the archive, the bandwidth and the proposal below are recomputed from it at every call. The Propulate framework (Taubert et al., 2023), which hands the whole history to each call and takes one candidate back, provides this contract (Appendix C).

3.3 Bandwidth (step 2)

The smooth kernel K_ϵ of (1) replaces the accept-or-reject threshold of classical ABC, and the bandwidth ϵ_n sets how sharply the archive weights and the posterior concentrate. The bandwidth tightens as the archive improves and never loosens, so that every worker, whatever prefix of the history it holds, is on the same tightening sequence. The bandwidth in force at call n is the smaller of the running minimum of the bandwidths stored in the history and the value a scheduler proposes from that history. The scheduler starts from a user-set ϵ_0 . At each search it proposes the largest bandwidth at which the kernel-weighted effective sample size of the archive retains 95% of its current value, by at most a factor of two, and it holds the bandwidth until $2k$ evaluated particles lie within it, so that ϵ_n settles near that order statistic of the discrepancies (settings in Appendix C). On an expensive simulator the run can end before the schedule has moved far from ϵ_0 , so ϵ_0 has to be set from the scale of the prior-predictive discrepancies (§5.3, §6.4).

3.4 Archive proposal (steps 3 to 5)

The proposal at call n is a Gaussian mixture over the archive,

$$q_n(\theta) = \sum_{j \in A_n} \tilde{W}_j^{(n)} K_\Sigma(\theta - \theta_j), \quad \tilde{W}_j^{(n)} \propto w_j \cdot K_{\epsilon_n}(\rho_j), \quad (4)$$

with $\sum_j \tilde{W}_j^{(n)} = 1$: the archive weight of a member is its parent weight times its kernel value at the current bandwidth. The perturbation kernel K_Σ is Gaussian with covariance $\Sigma_n = s \widehat{\text{Cov}}_{\tilde{W}}(A_n)$, the weighted covariance of the archive scaled by a constant s , and each component is truncated to the prior box. A candidate is drawn by choosing a parent in proportion to $\tilde{W}^{(n)}$ and perturbing it. Because the kernel is smooth, an archive member’s contribution shrinks continuously as the bandwidth tightens instead of dropping out when a threshold crosses it, as it does in hard-threshold ABC-PMC. The numerical details are in Appendix C.

3.5 Weights and the posterior estimate (step 6)

A candidate drawn from q_n should be importance-weighted against the proposal it came from, but the proposal moves with every arrival, so the history was drawn from a long sequence of proposals. AMIS resolves this by weighting against the mixture of the proposals used (§2). A *proposal snapshot* q_s is the proposal (4) rebuilt from the history truncated at call s , at the bandwidth stored there; any snapshot can be rebuilt from the history alone.

The parent weight. The running sampler keeps the last S snapshots it has taken, one every k calls. A candidate θ^* drawn from q_n receives

$$w^* = \frac{\pi(\theta^*)}{\bar{q}_n^{\text{on}}(\theta^*)}, \quad \bar{q}_n^{\text{on}}(\theta) = \frac{1}{1 + |\mathcal{S}|} \left[q_n(\theta) + \sum_{s \in \mathcal{S}} q_s(\theta) \right], \quad (5)$$

with $\mathcal{S}$ the kept snapshots; $S = 0$ recovers single-proposal weighting. This weight is stored with the particle as its w_i and enters the archive weights of (4) and the scheduler’s effective-sample-size search (§3.3), and nothing else. The kept snapshots are the only state the sampler carries between calls, and a stale or empty buffer changes which parents are chosen, not the estimate.

The posterior weights. After the run, the estimate rebuilds its own snapshots from $\mathcal{H}_n$, at most $S+1$ of them (two when $S = 0$), at history indices $\tau_1 < \dots < \tau_m$ spread in proportion to the number of draws each stands in for, and reweights every evaluated particle against their mixture. For a function f of the parameter the

estimate is

$$\hat{\pi}_n(f) = \frac{\sum_{i=1}^n W_{i,n} f(\theta_i)}{\sum_{i=1}^n W_{i,n}}, \quad W_{i,n} = \frac{\pi(\theta_i) K_{\epsilon_n}(\rho_i)}{\bar{q}_n(\theta_i)}, \quad (6)$$

with the denominator

$$\bar{q}_n = \lambda_n \pi + (1 - \lambda_n) \sum_{s=1}^m \alpha_{s,n} q_{\tau_s}, \quad \lambda_n = \max(\nu_n, \delta), \quad \alpha_{s,n} \geq 0, \quad \sum_s \alpha_{s,n} = 1. \quad (7)$$

This is the mixture of the m reconstructed snapshots, weighted in proportion to the draws each stands in for, and of the prior, whose weight λ_n is the share ν_n of prior draws in the history floored at a fixed $\delta > 0$; the prior component keeps the denominator bounded away from zero and so bounds every weight (Owen & Zhou, 2000). The snapshot mixture is a fixed-size stand-in for the mixture the sample was actually drawn from, and §4 asks how faithful a stand-in it is. Computing $\hat{\pi}_n$ is one $\mathcal{O}(nk)$ pass over the history, run after the timed budget and excluded from every throughput measurement (Appendix C).

3.6 Update loop

Algorithm 1 is the update a worker runs whenever it needs a candidate; its lines follow the steps of Fig. 1. On W workers each one runs the loop on its own copy of the history, which keeps growing as results arrive from every worker. No line waits for any other worker, and a worker that returns from a slow simulation finds a larger history than the one it left.

Algorithm 1 One update of the generation-free sampler, run by a worker on its copy of the history

Require: History $\mathcal{H}_n$, archive size k , starting bandwidth ϵ_0 , kept snapshots $\mathcal{S}$

- (1) **if** $|\mathcal{H}_n| < k$ **then return** a uniform prior draw with $w^* = 1$ ▷ bootstrap
 - (2) $A_n \leftarrow$ the k members of $\mathcal{H}_n$ with the smallest ρ_i ; $\epsilon_n \leftarrow$ the smaller of the stored running minimum and the scheduler’s proposal
 - (3) $\tilde{W}_j^{(n)} \propto w_j K_{\epsilon_n}(\rho_j)$ over A_n ; $\Sigma_n \leftarrow s \widehat{\text{Cov}}_{\tilde{W}}(A_n)$
 - (4) q_n as in (4); every k calls, add q_n to $\mathcal{S}$
 - (5) draw a parent $j \sim \tilde{W}^{(n)}$ and $\theta^* \sim K_{\Sigma_n}(\cdot - \theta_j)$, retired into the prior box
 - (6) w^* as in (5)
- return** θ^* for simulation; $(\theta^*, \rho^*, \epsilon_n, w^*)$ joins every worker’s history when ρ^* arrives
-

4 Theoretical Analysis

The estimate $\hat{\pi}_n$ of (6) divides each particle’s weight by $\bar{q}_n$, a fixed-size mixture of reconstructed proposals, whereas the particles were drawn from the sequence of all proposals the run used. Consistency therefore depends on how well $\bar{q}_n$ represents the density from which the pooled sample was drawn. Write $\tilde{q}_i$ for the conditional density of the i -th candidate given the past (the filtration of Assumption 4); it absorbs everything the implementation does, and it is the only way the sampler enters the theorem. The mixture the pooled sample was drawn from is

$$\bar{q}_n^* := \frac{1}{n} \sum_{i \leq n} \tilde{q}_i, \quad (8)$$

and the *fidelity ratio* r is the limit of $\bar{q}_n^*/\bar{q}_n$. We prove that the estimate converges to the smooth-ABC posterior tilted by r , so that $r \equiv 1$ gives the posterior itself. The proof is self-contained rather than a corollary of AMIS theory, because the published consistency results for recycling schemes (Marin et al., 2019) restrict the adaptation so that the weighting stage decouples, which a single-arrival scheme does not do. A central limit theorem for variants whose adaptation is damped, with a per-run variance estimator, and the decomposition and partial measurement of r are in Appendix A; the proofs are in Appendix B. Throughout, ϵ_∞ is the limit of the bandwidth sequence, whose existence and positivity are part of Assumption 5(i), and $L_\epsilon(\theta) = \mathbb{E}[K_\epsilon(\rho) \mid \theta]$ is the smooth ABC likelihood of (1).

Assumptions.

Assumption 1 (kernel). $K_\epsilon : [0, \infty) \rightarrow [0, 1]$ is measurable with $K_\epsilon(0) = 1$ (unit *peak*, not unit integral, as is standard for a probabilistic-acceptance ABC kernel), and for every $a \in (0, \epsilon_0]$ there is $L_K(a) < \infty$ with $|K_\epsilon(\rho) - K_{\epsilon'}(\rho)| \leq L_K(a)|\epsilon - \epsilon'|$ for all $\rho \geq 0$ and $\epsilon, \epsilon' \in [a, \epsilon_0]$. The limiting target is non-degenerate: $Z_{\epsilon_\infty} = \int_\Theta \pi L_{\epsilon_\infty} > 0$.

Assumption 2 (denominator class). $\Theta \subset \mathbb{R}^d$ is a non-degenerate axis-aligned box and the prior density satisfies $0 < \pi_- \leq \pi \leq \pi_+ < \infty$ on it. Each q_{τ_s} in (7) is a positive function on Θ of truncated-Gaussian-mixture form: at most k components, means in Θ , weights in the simplex, covariance Σ with $\sigma_-^2 I \preceq \Sigma \preceq \sigma_+^2 I$ for fixed $0 < \sigma_- \leq \sigma_+ < \infty$, and per-component normalizing constants evaluated as the implementation evaluates them.

Assumption 3 (prior floor in the denominator). $\delta \in (0, 1/2]$ is fixed and $\lambda_n \geq \delta$ in (7).

Assumption 4 (adapted filtration). There is a filtration $(\mathcal{F}_i)$ with $\xi_i = (\theta_i, \rho_i)$ $\mathcal{F}_i$ -measurable, $\theta_i \mid \mathcal{F}_{i-1} \sim \tilde{q}_i$ and $\rho_i \mid \mathcal{F}_{i-1} \vee \sigma(\theta_i) \sim p(\cdot \mid \theta_i)$. Under sequential execution this holds by construction; under asynchronous execution it is an assumption (Appendix B).

Assumption 5 (fidelity and stabilization). (*i, consistency*) $\epsilon_n \downarrow \epsilon_\infty > 0$ almost surely (ϵ_∞ may be random), and the denominator tracks the draw mixture up to a limiting ratio: for some r with $0 < r_- \leq r \leq r_+ < \infty$,

$$\|\bar{q}_n^*/\bar{q}_n - r\|_\infty \longrightarrow 0 \quad \text{a.s.} \quad (9)$$

(*ii, central limit theorem*) A stabilization rate and deterministic limits, needed only for Theorem 3 and stated in Appendix A.

Assumption 1 holds for the Gaussian kernel with $L_K(a) = 2/(ea)$ and Epanechnikov with $2/a$; it excludes the hard indicator, which is not Lipschitz in ϵ . This is one reason the method uses smooth kernels. Assumptions 2 and 3 constrain only the m -snapshot denominator $\bar{q}_n$, which the implementation builds to that exact recipe. Assumption 5 holds the remaining conditions. For consistency, (9) is the whole of the sampler's contribution: the sampler need not take any particular form, but whatever it does must be asymptotically reflected in the denominator, up to a ratio r that the results carry. The central limit theorem needs in addition the per-draw stabilization rate and the deterministic limits of clause (ii).

Theorem 1 (consistency). *Let Assumptions 1–4 and 5(i) hold. Then for every bounded measurable $f : \Theta \rightarrow \mathbb{R}$,*

$$\hat{\pi}_n(f) \longrightarrow \pi_{\epsilon_\infty}^r(f) := \frac{\int_\Theta f \pi L_{\epsilon_\infty} r}{\int_\Theta \pi L_{\epsilon_\infty} r} \quad \text{almost surely.} \quad (10)$$

Corollary 2 (exact fidelity). *If $r \equiv 1$, that is, if the denominator is asymptotically the density the sample was drawn from, then $\hat{\pi}_n(f) \rightarrow \pi_{\epsilon_\infty}(f)$, the smooth-ABC posterior at the limiting bandwidth.*

What r is, and how much of it is measured. Everything specific to the implementation enters through r , in four factors: the prior floor δ in the denominator, the m reconstructed snapshots standing in for the n proposals actually used, the approximate normalization of the truncated mixture components, and the sampler's departures from drawing exactly from q_n (Appendix A.1). The first two can be estimated from a stored run by recomputing the denominator at a much larger m and at the observed prior share: on the five Cellular Potts production runs the estimate lies 0.2–0.5% in total variation from that reference, and on a one-dimensional Gaussian history 0.26%. The other two are bounded, not measured, so the accounting of r is partial and the results are conditional on its finite positive limit (Table 5). The experiments use $S = 20$, so $m \leq 21$, and the floor is $\delta = 0.5/(m+1)$ (Appendix C).

5 Experimental Design

5.1 Benchmarks

We evaluate calibration and recovery on targets with reference posteriors, test the barrier prediction under runtime laws known by construction, and measure performance on a real simulator of the class described

Table 1: Benchmark suite. Cost is the mean wall-clock time of one simulation in the asynchronous runs (an evaluation on Cellular Potts is four replicate simulations); runtime spread is the in-run coefficient of variation of that time, or how it is injected. Budgets are per replicate; five replicates throughout except three at 80^3 .

Benchmark	d	cost/sim	runtime spread	budget, workers	role
Gaussian mean	1	0.1 ms	injected: straggler 0–20 $\times$; lognormal $\sigma \leq 2$	300 s, 48 (16 in the straggler study)	calibration; barrier cost under a known runtime law; the cheap end of the cost boundary
g-and-k	4	2 ms	none	600 s, 48	4-D calibration and recovery against a reference posterior
Lotka–Volterra	4	4 ms	trajectory-dependent; 98% extinct	600 s, 48; scaling to 288	strong scaling; the cost boundary
Cellular Potts 50^3	2	13 s	CV 0.05	3600 s, 48; scaling to 384	posterior recovery against a known truth; barrier cost on a real simulator
Cellular Potts 80^3	2	170 s	CV 0.13–0.23	3600 s, 48	barrier cost at realistic cost and heterogeneity

in §2. Table 1 lists the five configurations. The one-dimensional Gaussian mean has an analytic posterior and the four-dimensional g-and-k distribution (Rayner & MacGillivray, 2002) a reference posterior; both are near-instant simulators, so they also expose the cost of coordinating one arrival at a time. The two studies with injected runtime heterogeneity run on the Gaussian mean: a persistently slow worker, and a lognormal multiplier on every evaluation’s runtime. Lotka–Volterra is a stochastic simulator whose cost varies with the trajectory and whose parameters send 98% of simulations to extinction (Appendix D); it serves the scaling and boundary measurements and is not scored for posterior quality.

The Cellular Potts benchmark is the real simulator, and its runtime spread is a property of the inference problem rather than of the machine. How fast cells divide and how large they grow determine how many cells the lattice holds, and the cell count fixes both the summaries and the runtime (correlation 0.99 between runtime and cell count). The spread grows with the lattice, because at 80^3 the box no longer bounds the cell count. We run the `cellsInSilico` implementation (Berghoff et al., 2020) of the three-dimensional Cellular Potts model (Graner & Glazier, 1992) end to end at two lattice sizes, and infer the two parameters the summary statistics resolve, the cell division rate and the target cell volume, from the final cell count and the cluster radius; one evaluation averages four simulations, because a single realization is too noisy to order nearby parameters. The simulations take seconds to minutes rather than the hours of a production study, and the observations are synthetic with a known truth, so the benchmark scores both the systems claims and posterior recovery. The 50^3 twin and scaling runs used an earlier parameterization of the same simulator (division rate and motility, one simulation per evaluation); only its runtime distribution enters those results. Appendix E gives the screening experiment that fixed the parameters and summaries, the prior ranges, and both configurations.

5.2 Baselines and instruments

We use three comparators, each for a different question. *pyABC* is pyABC (Klinger et al., 2018; Schälte et al., 2022) with its acceptor replaced by a probabilistic-rejection acceptor using the same smooth kernel and the same bandwidth schedule as the asynchronous sampler (Appendix C); it is the method a practitioner would run, and the comparator for C2 and C3. Its population of 100 on 47 simulating workers (one process only dispatches) forces at least three rounds per generation, the last one 13% full, so it idles by construction even on a homogeneous simulator. The scaling study gives it a population equal to the worker count; the production comparisons do not. The *twin* is our own propagator with a collective barrier inserted before each proposal and nothing else changed. Histories are not truncated to generation boundaries, so each worker proposes from the history it has received at the barrier. The twin thus isolates the timing of synchronization from every other difference, and it is the comparator for C1. A collective barrier needs identical call counts on every worker, so the twin runs to a fixed simulation count, the asynchronous sampler runs to a fixed wall clock, and the two are compared as rates. For the posterior comparison the asynchronous sampler is also

re-run at exactly the twin’s simulation count and seeds. *Rejection ABC* spends the same simulation budget, keeps the k best draws, and is given the whole machine (all 48 workers) rather than one worker per replicate. It is the reference for the per-simulation gain of an adaptive proposal. All three run over MPI on JUWELS (Appendix D) and are samplers of the same estimator class; a neural simulation-based-inference baseline would change the estimator rather than the barrier, and its rounds would wait for their slowest simulation as generations do (§2).

5.3 Metrics

Tolerance at a matched budget. To compare samplers on one footing we use the k -th order statistic of a method’s own discrepancies after n simulations, $\epsilon_{(k)}(n)$: the tolerance at which exactly k of its draws would be accepted. It needs no reference posterior, it does not depend on the bandwidth a schedule happened to report at, and it is what rejection ABC reports at by construction. The *per-simulation efficiency* of the asynchronous sampler over pyABC is the ratio of their $\epsilon_{(k)}$ at the same n , above one when the asynchronous tolerance is the tighter; the *equal-wall-clock* ratio is the same quantity at the end of both runs, and the two differ by the throughput ratio. The slope of $\epsilon_{(k)}(n)$ in $\log n$ shows whether more simulations keep converting into a tighter tolerance. We use $k = 100$ throughout, over completed simulations only, ordered by completion time (Appendix D).

Throughput, utilization, and the predicted barrier cost. Throughput is simulations completed per second of wall clock; utilization is the fraction of worker time spent inside the simulator. The prediction that C1 tests is built from the asynchronous run only. Let D_i be the per-evaluation durations on that run (recorded simulation time plus the measured per-evaluation coordination overhead). A barrierized generation of W evaluations takes $\mathbb{E}[\max_{i \leq W} D_i]$, so the synchronous throughput is predicted as $T_{\text{sync}} = W/\mathbb{E}[\max_{i \leq W} D_i]$. The expectation is taken by resampling W durations from the asynchronous record, or from the injected law where the heterogeneity is injected; a persistently slow worker contributes its own mean. The predicted cost is $T_{\text{async}}/T_{\text{sync}}$ with T_{async} the measured asynchronous throughput, and it is compared with the ratio measured against the twin. For identically distributed, uncensored durations $T_{\text{async}} = W/\mathbb{E}[D]$ and the ratio is the straggler factor of §1. With a persistently slow worker, or when a wall-clock budget censors the long durations, the two differ, and we use the general form. The prediction has a domain: the simulation time must dominate the latency of the collective itself, which the model does not contain, and the timing sample must be long enough to cover the tail of the runtime distribution. Configurations outside this domain are marked as such in §6.1.

Posterior quality. Calibration is checked by simulation-based calibration (Talts et al., 2018): over 1000 trials, each with a parameter drawn from the prior and data simulated from it, the empirical coverage of the posterior’s central intervals at the 0.50/0.80/0.90/0.95 levels. The effective sample size (ESS) of a weighted posterior is $(\sum_i W_i)^2 / \sum_i W_i^2$. Where a reference posterior exists we report the Wasserstein distance W_1 between each method’s own posterior and that reference (in more than one dimension the sliced W_1 over random projections (Bonneel et al., 2015), on the parameters’ natural scale). The reference is the analytic posterior for the Gaussian mean; for g-and-k it is the posterior given the same octile summaries the discrepancy is built from, obtained from their asymptotic law (Appendix D). The asynchronous estimator is replayed over the prefix of the history completed by each checkpoint. pyABC is scored on its last completed generation, with pyABC’s population weights where the history was kept and uniform weights otherwise, and rejection ABC on its k best draws. On Cellular Potts, whose truth is known but whose posterior is not, we report each parameter’s *contraction* $1 - \text{sd}/\text{sd}_{\text{prior}}$ in prior-normalized units, whether the central 90% interval covers the truth, and the ESS of the posterior weights.

Starting bandwidth. On an expensive simulator the bandwidth schedule may not move far from ϵ_0 within the budget (§3.3), so ϵ_0 is set before the run from the discrepancies of a few thousand prior-predictive draws, at about a fifth of their median; §6.4 measures what a starting value far above that costs.

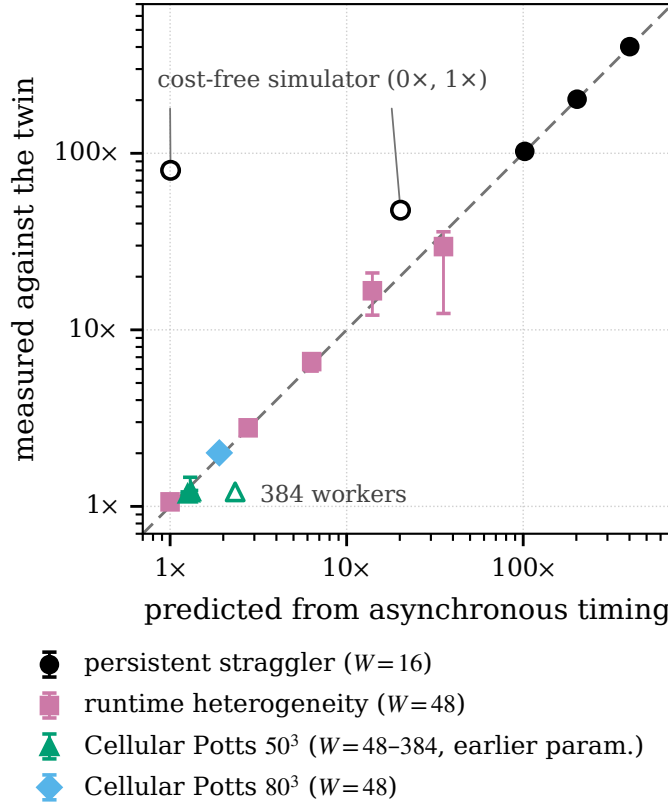


Figure 2: The barrier’s cost predicted from the asynchronous run’s timing alone (§5.3) against the cost measured with the barrierized twin, on every configuration the twin was run on: a persistent straggler at five slowdowns, lognormal runtime heterogeneity at five spreads, and Cellular Potts at 50^3 (48–384 workers, earlier parameterization, utilization ratio) and 80^3 (48 workers, against pyABC). Medians with replicate ranges; the dashed line is equality; open markers are the three configurations outside the predictor’s domain.

6 Results

Unless noted, results are medians over five replicates with the replicate range shown; *asynchronous* is the proposed sampler, *pyABC* the baseline with the same kernel and schedule, and *twin* the barrierized copy of our own propagator (§5.2).

6.1 The barrier costs the straggler factor, and the cost is predictable (C1)

The prediction holds. Figure 2 covers every configuration on which we ran the barrierized twin: a persistently slow worker at five slowdowns, lognormal runtime heterogeneity at five spreads, and the Cellular Potts simulator at four worker counts and two lattice sizes. For each, the barrier’s cost as predicted from the asynchronous run’s timing alone (§5.3) is plotted against the cost measured with the twin. Over the twelve configurations inside the predictor’s domain the predicted-to-measured ratio runs from 0.84 to 1.19 with median 0.996, across measured costs from $1.2\times$ to $400\times$ (Tables 9, 10 and 11 in Appendix F). The two largest misses, 16% and 19% at the widest runtime spreads, occur where the twin completes only ten generations per replicate and the 60s budget censors the asynchronous run’s longest draws. Three configurations lie outside the domain and are drawn open: the cost-free simulator at $0\times$ and $1\times$, where the collective’s own latency, about 0.3s per barrier, dominates a 4ms workload; and 384 Cellular Potts workers, where the asynchronous timing sample carries a contention tail that the twin’s generations did not show.

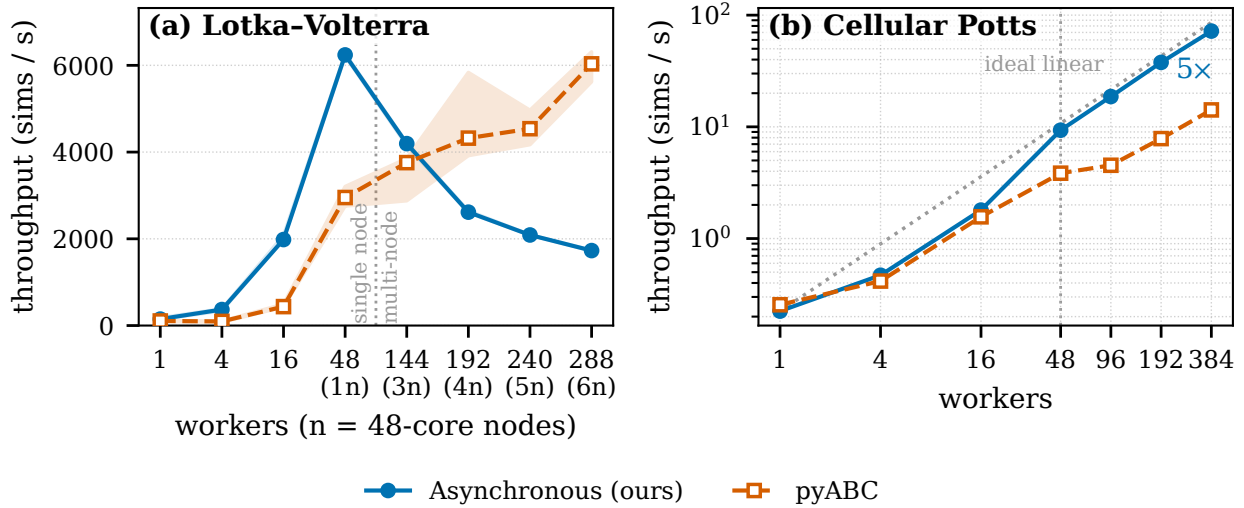


Figure 3: Strong scaling at a fixed wall-clock budget against pyABC with population $\max(100, W)$. **Left:** Lotka–Volterra (4ms; 180s; medians and inter-quartile ranges over five replicates). **Right:** Cellular Potts 50^3 (earlier parameterization, 5.1s; 1800s; log–log), with linear scaling dotted.

What the twin does not isolate. On Cellular Potts at 50^3 the twin’s throughput was 1.9–2.8 $\times$ below the asynchronous sampler’s, but only part of that is the barrier. Throughput is utilization over mean simulation time, and the two factors separate: the utilization ratio is 1.2 $\times$ at every worker count and matches the prediction, while the rest is the twin’s simulations running longer for reasons unrelated to the parameters, which we do not attribute (Table 11). The barrier’s cost on this workload is therefore 1.2 $\times$, as its runtime spread predicts, and it rises with the lattice: at 80^3 , where an evaluation costs 170s and the runtime coefficient of variation is 0.23, the predicted cost is 1.9 $\times$ and the gap measured against pyABC 2.1 $\times$. The same comparison against pyABC at 50^3 gives 2.3–2.4 $\times$, of which only 1.2 $\times$ is the barrier; the rest is pyABC’s fixed per-generation overhead, a large fraction of a 13s evaluation and negligible against a 170s one. We therefore quote 2.0 $\times$ as the Cellular Potts systems gain attributable to the barrier. The asynchronous sampler itself scales almost linearly on this workload, at 97% parallel efficiency from one to eight nodes (Fig. 3, right).

Against pyABC. pyABC is what a practitioner would run instead, and the same experiments score it (Fig. 8, Appendix F). Under the persistent straggler its throughput falls more than a hundredfold while the asynchronous sampler’s falls by less than half, because every generation waits for the slow worker. Under runtime heterogeneity its workers idle 51–79% of their time against 1–27% for ours, and its throughput is 2.1–4.8 $\times$ below ours. That is less than the twin loses, because pyABC closes a generation on the first 100 finishers and discards the latecomers it has already paid for, so the comparison a practitioner faces understates the barrier’s full cost. On Cellular Potts pyABC runs at 43–49% utilization against 98–99.8%, and the asynchronous sampler completes 2.1–2.4 $\times$ its simulations in the same budget.

6.2 Throughput converts into a tighter tolerance (C2)

Throughput matters only through the tolerance a run reaches. Table 2 reads it off every benchmark as $\epsilon_{(100)}(n)$ (curves in Fig. 9, Appendix F). At pyABC’s own simulation count the asynchronous sampler is the more efficient per simulation on three of the four benchmarks; the one on which it is worse is the one-dimensional analytic target. Both the per-simulation ratio and the throughput ratio rise with the cost of a simulation, and their product, the ratio at equal wall clock, exceeds one on Lotka–Volterra and reaches 4.1 $\times$ on Cellular Potts. On the two cheap benchmarks the asynchronous and pyABC curves have the same slope, so the per-simulation ratio is a constant factor; on Cellular Potts both adaptive samplers steepen well

Table 2: Tolerance at a matched budget on every benchmark ($k = 100$ order statistic of each method’s own discrepancies; five replicates; medians, with the replicate range of each ratio in parentheses; completed simulations only). Per-simulation: ϵ of pyABC over the asynchronous sampler at pyABC’s own simulation count; equal wall clock: the same ratio at the end of both runs.

Benchmark	cost/sim	sims (async / sync)	throughput	per-simulation	equal wall clock
Gaussian mean	0.1 ms	1.31M / 4.61M	0.28×	0.78× (0.77–1.12)	0.24× (0.21–0.28)
g-and-k	2 ms	1.75M / 3.81M	0.46×	1.06× (1.02–1.15)	0.88× (0.85–0.92)
Lotka–Volterra	4 ms	1.51M / 1.56M	0.96×	1.30× (1.08–1.51)	1.30× (1.02–1.43)
Cellular Potts	13 s	12,900 / 5,400	2.39×	1.6× (1.14–1.94)	4.09× (3.1–6.6)

beyond the n^{-1} of rejection ABC in this noise-dominated regime, so their advantage over it widens with every doubling of the budget.

Cellular Potts, fixed production run. Table 4 gives the decomposition on the production run. In the same hour the asynchronous sampler completes 2.4× the simulations of pyABC, of which §6.1 attributes 1.2× to the barrier at this lattice size; it is 1.6× more efficient per simulation, read at pyABC’s count of 5,400 where pyABC’s curve is the steeper (Fig. 9d), and ends 4.1× tighter at equal wall clock. A rejection sampler given the same budget and the whole machine ends 39× looser. At 80³ the asynchronous sampler completes 2.1× the simulations and is 2.1× more efficient per simulation.

6.3 Where the advantage turns on (C3)

Each arrival incurs a roughly fixed coordination cost: rebuilding the $\mathcal{O}(k)$ proposal, weighting the arrival against the kept snapshots, and exchanging it with every other worker. This cost becomes less important as the simulation cost increases. Figure 4 plots the throughput and per-simulation ratios of Table 2 against the cost of one simulation. Both ratios increase with simulation cost. Only the throughput ratio has a common mechanism across benchmarks; the per-simulation ratio also depends on dimension, summaries and noise in the four inference problems. We therefore base the boundary on throughput: parity is reached at about 4 ms per simulation, and the net advantage at equal wall clock turns positive between 2 and 4 ms. That boundary belongs to the placement and budget it was measured under: the cheap benchmarks ran 48 workers spread over four nodes (Appendix D), so every arrival crosses the network, and on one packed node the same Lotka–Volterra configuration gives a throughput ratio of 2.1 instead of 0.96 (Table 12). Below the boundary a barrier costs nothing worth removing: on the 0.1 ms Gaussian mean pyABC completes 3.5× more simulations and ends with a 4× tighter tolerance at equal wall clock.

The same per-arrival cost sets the scaling boundary on a cheap simulator. On Lotka–Volterra the exchange of every arrival with every worker grows with the worker count, so a worker is inside the simulator 25% of its time at 48 workers and 2% at six nodes; throughput peaks on one node, and pyABC with a population equal to the worker count overtakes beyond three nodes (Fig. 3, left). At 13 s per evaluation the same coordination is negligible, and the method tracks linear scaling to eight nodes (Fig. 3, right).

6.4 The posterior: calibration, recovery, and what limits it (C4)

Calibration. Table 3 gives simulation-based calibration of the full-history estimator (1000 trials, matched simulation budget). On the one-dimensional Gaussian mean it tracks the nominal coverage at every level, while pyABC under-covers at every level (its nominal 0.90 interval covers 0.75); the Monte Carlo error of ± 0.01 resolves this ordering. On a bimodal target ($y_i \sim \mathcal{N}(\theta^2, \sigma^2)$, the case a top- k archive is most at risk of collapsing onto one mode) the archive retains both modes in 86% of trials and coverage is nominal. On the four-dimensional g-and-k model the estimator is *conservative* on A and g and near nominal on B and k , with B ’s rank histogram the only uniform one at the 1% level. pyABC is near nominal on A and g and under-covers B and k , with no uniform rank histogram. The evidence points to a reporting-support effect rather than a weighting failure: the top- k archive alone would under-cover (0.85/0.89 at the 0.90/0.95 levels), the full history over-covers, and coverage rises monotonically through nominal as the support grows from one

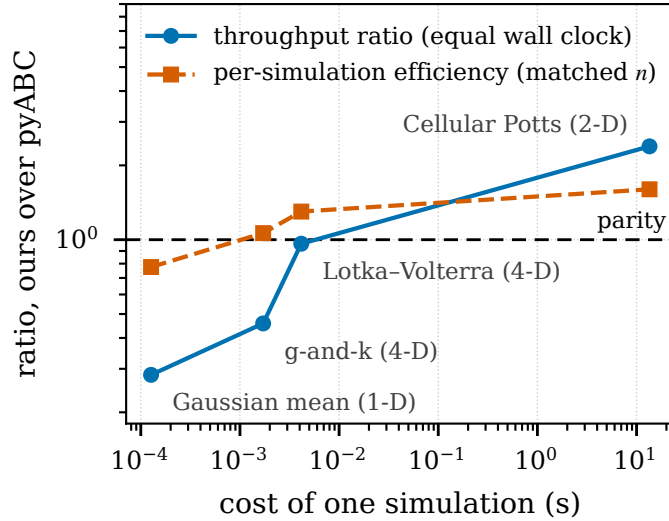


Figure 4: Throughput ratio and per-simulation efficiency of the asynchronous sampler over pyABC against the cost of one simulation, one point per benchmark (Table 2).

Table 3: Simulation-based calibration of the posterior estimate (1000 trials; empirical coverage of central intervals at the nominal levels; Monte Carlo standard error ± 0.007 – 0.016). Nominal is best.

Target	Method	0.50	0.80	0.90	0.95
Gaussian mean (1-D)	asynchronous	0.51	0.80	0.90	0.94
	pyABC	0.35	0.63	0.75	0.83
Bimodal $\mathcal{N}(\theta^2, \sigma^2)$ (1-D)	asynchronous	0.50	0.82	0.91	0.96
	pyABC	0.45	0.82	0.90	0.95
g-and-k (4-D), A	asynchronous, full history	0.62	0.91	0.96	0.99
	pyABC	0.54	0.82	0.91	0.95
g-and-k (4-D), B	asynchronous, full history	0.54	0.83	0.91	0.96
	pyABC	0.45	0.74	0.85	0.90
g-and-k (4-D), g	asynchronous, full history	0.58	0.88	0.95	0.98
	pyABC	0.51	0.82	0.91	0.95
g-and-k (4-D), k	asynchronous, full history	0.56	0.83	0.92	0.96
	pyABC	0.47	0.74	0.86	0.92
g-and-k (4-D), A	asynchronous, top- k archive	—	—	0.85	0.89

to the other. Doubling the simulation budget leaves the coverage unchanged. A sweep over the archive size and the number of kept snapshots on a d -dimensional Gaussian target (Appendix F) finds a miscalibrated region at $d = 8$, $k = 50$; of the archive sizes swept, $k = 100$ is the best calibrated.

Recovery against a reference posterior. Where a reference exists, Fig. 5 scores each method’s posterior against it over wall-clock prefixes. On the Gaussian mean the asynchronous estimator is closer to the analytic posterior than pyABC’s final population at every checkpoint, and a fairly budgeted rejection sampler is as close; in one dimension with that many draws, rejection is enough. On the four-dimensional g-and-k the order reverses: pyABC ends at $W_1 = 0.029$ and the asynchronous estimator at 0.069, with rejection far behind at 0.47. The effective sample sizes explain this reversal. The posterior’s effective sample size stays at 238–296 at every checkpoint of the Gaussian run while the history grows from 10^5 to 1.3×10^6 particles. A controlled sweep puts ESS/k between 1.9 and 3.6 for $k \in \{50, 100, 200\}$. The estimate is reported at the tightest bandwidth reached, which is chosen by ESS retention against the archive (§3.3). Its effective support therefore stays at a few multiples of k however many particles are evaluated. In one dimension that

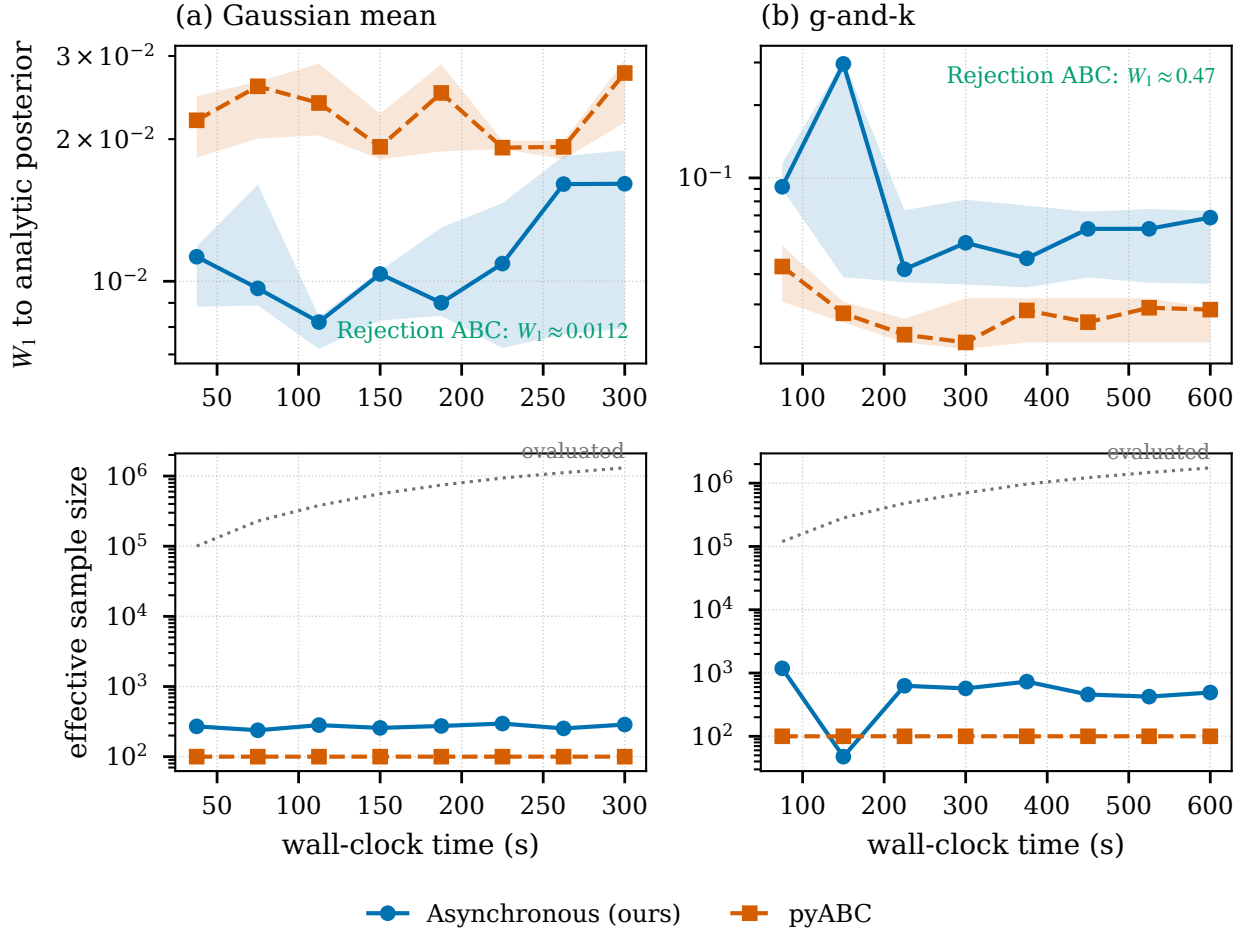


Figure 5: Each method’s own posterior (§5.3) scored against the analytic posterior (a) and the g-and-k reference posterior (b) over wall-clock prefixes. Top: W_1 , median and inter-quartile range over five replicates. Bottom: the effective sample size of the posterior estimate, with the evaluated-particle count dotted.

support is small and stable; in four it is small and erratic (7 to 2987 across checkpoints and replicates), and accuracy follows it. The four-dimensional deficit is consistent with intermittent weight degeneracy rather than bias. The limitation appears once the target is identified well enough for the bandwidth to tighten.

A known truth on the real simulator. On the two-parameter Cellular Potts setup (one synthetic dataset with a known truth), all three methods recover the truth (Fig. 6, Table 4). The division rate is well identified by every method (91–92% contraction against the prior). For cell volume, contraction is $81\% \pm 1\%$ for the asynchronous estimator, $72\% \pm 4\%$ for pyABC’s last generation with its population weights, and 67% for the best 100 rejection draws. The truth lies inside the central 90% interval of every run (five per adaptive method, one rejection run); on a single dataset that is containment, not coverage. The asynchronous estimator is the tightest and has the largest effective sample (303 against pyABC’s 66 of 100, medians).

The starting bandwidth. The first row of Table 4 is the same sampler started at the benchmark’s original $\epsilon_0 = 10$, twenty times the prior-predictive median discrepancy of 0.46. It reports $13\% \pm 14\%$ contraction on the cell volume, erratic as well as poor, although its draws are as concentrated as the production run’s (the end tolerances agree within 20%). The runs differ in the bandwidth used for reporting. After the run’s 62 calls per worker, the schedule had not moved from ϵ_0 , which ended $321\times$ above the tolerance the draws supported, against $2\times$ at $\epsilon_0 = 0.1$. The transient is negligible on a simulator that affords millions

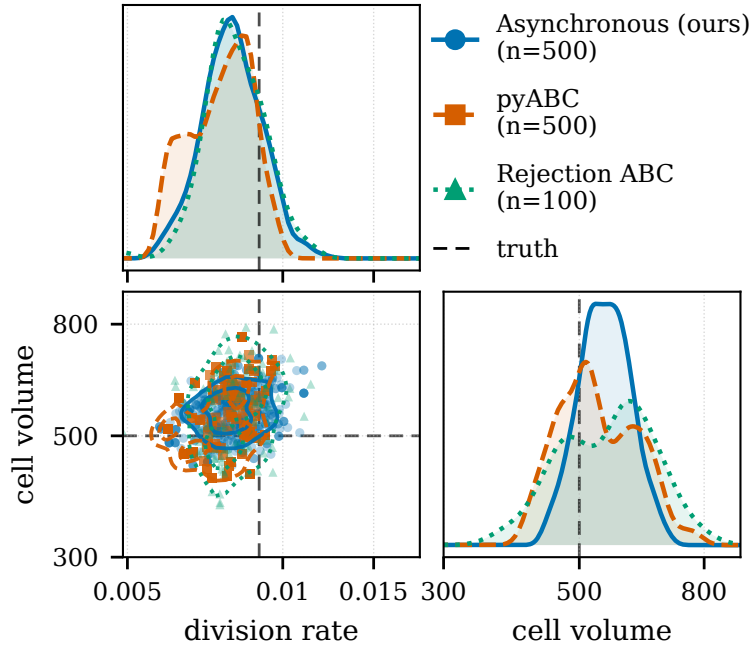


Figure 6: Cellular Potts posterior (division rate, cell volume) on the fixed production run, replicate 0: the asynchronous AMIS estimator (500 resampled draws), pyABC’s last completed generation with its population weights (500 resampled draws), and the 100 best rejection draws (Table 4). Axes are the log-uniform prior coordinates ticked in physical units; dashed lines mark the truth.

Table 4: Cellular Potts 50^3 , five replicates per adaptive method and one rejection run, 3600s on 48 workers (means $\pm$ s.d. for contraction, medians otherwise). The first row is the asynchronous sampler at the benchmark’s original starting bandwidth (10, twenty times the prior-predictive median discrepancy), the rest at the production value (0.1). Rejection ABC is one run of 13,000 prior draws on all 48 workers, keeping the best 100. Contraction is $1 - \text{sd}/\text{sd}_{\text{prior}}$; the truth lies inside the central 90% interval of every run.

	sims	util.	$\epsilon_{(100)}, \text{end}$	contraction		ESS
				division rate	cell volume	
Asynchronous, $\epsilon_0 = 10$	13,300	99.8%	6.3×10^{-5}	$84 \pm 6\%$	$13 \pm 14\%$	645
Asynchronous, $\epsilon_0 = 0.1$	12,900	99.8%	7.6×10^{-5}	$92 \pm 1\%$	$81 \pm 1\%$	303
pyABC, $\epsilon_0 = 0.1$	5,400	42.8%	3.1×10^{-4}	$91 \pm 1\%$	$72 \pm 4\%$	66
Rejection ABC, best 100 of 13,000	13,000	—	2.9×10^{-3}	91%	67%	100

of evaluations, which is why the cheap runs never showed it. It remains consequential in the regime the method is designed for. The rule of §5.3, about a fifth of the prior-predictive median, recovers the posterior, and re-reporting a stored history at the k -th order statistic of its own discrepancies, at no simulation cost, reaches a comparable posterior (93%/85% on replicate 0) and is the check we apply. The barrierized twin shows the same transient on the cheap straggler benchmark (Appendix F.2).

The archive size. The effective sample size is a few multiples of k , but raising k has costs. Re-reporting one history at the order statistic for k from 10 to 1000 loosens the final bandwidth and lowers the cell-volume contraction from 85% to 78%. Running with $k = 300$ leaves the schedule at ϵ_0 for the whole hour. On the cheap benchmarks, $k = 100$ is Pareto-optimal for throughput and calibration (Appendix F). The archive size is a benchmark-dependent choice, made by checking calibration and the bandwidth reached.

7 Discussion

Against pyABC with the same kernel and schedule, the method is ahead on the tissue simulator in simulations per hour, tolerance at equal wall clock, contraction on the weakly identified parameter and effective sample size (Table 4); on the synthetic targets it is calibrated wherever pyABC under-covers, and it loses only where a simulation costs less than a few milliseconds. The results give a practitioner four rules.

Use it above a few milliseconds per simulation. Below that the per-arrival coordination costs more than a barrier does, and a generation-based sampler with a population sized to the machine is the better tool; above it, removing the barrier recovers a cost that grows with the worker count and with the spread of simulation times (§6.3).

Predict the gain before paying for it. The barrier’s cost is $W/\mathbb{E}[\max_{i \leq W} D_i]$ against the asynchronous throughput, computable from the run’s own per-evaluation timing, with two cautions: the timing sample must be longer than the tail of the runtime distribution, and a persistently slow worker contributes its own evaluation time rather than its wall-clock span (§6.1).

Set the starting bandwidth from the prior-predictive discrepancy scale. On an expensive simulator the whole run can be spent inside the bandwidth transient, and a starting value twenty times the prior-predictive median cost most of the contraction on the weakly identified parameter; about a fifth of that median transferred, and re-reporting a stored history at the order statistic of its own discrepancies is the check (§6.4).

Keep the archive small and check calibration. The effective sample size is a few multiples of the archive size, and enlarging the archive loosens the final bandwidth and lengthens the transient; $k = 100$ was never far from the best choice on any benchmark, and the calibration checks are cheap to repeat (§6.4).

The effective-sample-size ceiling is where the method can still improve. Barrier removal delivers simulations predictably; converting all of them into posterior accuracy is limited by the reporting rule, which we adopted for its simplicity: report at the tightest bandwidth reached, chosen by ESS retention against the archive. Further work on this method belongs in that rule rather than in the scheduler.

8 Limitations

Consistency holds up to the fidelity ratio r , of which two factors are measured and two bounded. The adapted filtration under asynchronous execution is assumed (§4). The central limit theorem is proved for damped-adaptation variants, and the production runs sit at the threshold of its rate condition rather than clearly inside it (Appendix A). Keeping every evaluated particle over-represents parameter regions that simulate quickly, the length bias of deadline-limited sampling (Murray et al., 2021); a runtime-coupled study found no measurable shift of the posterior mean at the couplings tested, and we do not correct for it (Appendix F).

9 Conclusion

A generation barrier costs the ratio of the barrier-free throughput to what a generation allows, and that cost can be predicted from the asynchronous run’s timing alone: the prediction matched a barrierized twin of our sampler on synthetic runtime laws and on a real tissue simulator. The generation-free sampler we propose removes the barrier and, wherever a simulation costs more than a few milliseconds, converts the simulations it gains into a tighter tolerance; on the tissue simulator that is $2.4\times$ the simulations of pyABC in the same hour and a $4.1\times$ tighter tolerance. Its estimator is consistent up to a partly measured fidelity ratio, calibrated in one dimension and conservative in four, and replayable from the log. Two settings bound what it delivers: the starting bandwidth, which must be set from the prior-predictive scale, and the archive size, which sets the effective sample size.

Broader Impact Statement

This work is methodology for likelihood-free inference on parallel machines. Its direct effect is on compute: it removes the idle time a generation barrier imposes on heterogeneous simulator workloads, and the prediction

of §6.1 lets a practitioner estimate that saving from their own timing data before spending an allocation. We see no risk specific to the method beyond those of Bayesian inference in general.

Reproducibility

The repository accompanying this paper contains the summary data behind every figure and table, the scripts that regenerate each from the campaign output, and the committed output of every diagnostic quoted in the text, so every figure, table and quoted number can be regenerated from the repository alone. The full implementation (the Propulate propagator, the matched-kernel pyABC baseline, the analysis and plotting scripts, and the test suite), the raw per-particle records (≈ 30 GB for the Gaussian-mean campaign alone), the per-rank timing logs and the resolved run configurations will be released publicly upon publication.

References

- Emad Alamoudi, Felipe Reck, Nils Bundgaard, Frederik Graw, Lutz Brusch, Jan Hasenauer, and Yannik Schälte. A wall-time minimizing parallelization strategy for approximate Bayesian computation. *PLOS ONE*, 19(2):e0294015, 2024. doi: 10.1371/journal.pone.0294015.
- Mark A. Beaumont. Approximate Bayesian computation. *Annual Review of Statistics and Its Application*, 6:379–403, 2019. doi: 10.1146/annurev-statistics-030718-105212.
- Mark A. Beaumont, Jean-Marie Cornuet, Jean-Michel Marin, and Christian P. Robert. Adaptive approximate Bayesian computation. *Biometrika*, 96(4):983–990, 2009. doi: 10.1093/biomet/asp052.
- Marco Berghoff, Ivan Kondov, and Johannes Hötzer. Massively parallel stencil code solver with autonomous adaptive block distribution. *IEEE Transactions on Parallel and Distributed Systems*, 29(10):2282–2296, 2018. doi: 10.1109/TPDS.2018.2819672.
- Marco Berghoff, Jakob Rosenbauer, Felix Hoffmann, and Alexander Schug. Cells in Silico – introducing a high-performance framework for large-scale tissue modeling. *BMC Bioinformatics*, 21:436, 2020. doi: 10.1186/s12859-020-03728-7.
- Nicolas Bonneel, Julien Rabin, Gabriel Peyré, and Hanspeter Pfister. Sliced and Radon Wasserstein barycenters of measures. *Journal of Mathematical Imaging and Vision*, 51(1):22–45, 2015. doi: 10.1007/s10851-014-0506-3.
- Mónica F. Bugallo, Víctor Elvira, Luca Martino, David Luengo, Joaquín Míguez, and Petar M. Djurić. Adaptive importance sampling: The past, the present, and the future. *IEEE Signal Processing Magazine*, 34(4):60–79, 2017. doi: 10.1109/MSP.2017.2699226.
- Olivier Cappé, Arnaud Guillin, Jean-Michel Marin, and Christian P. Robert. Population Monte Carlo. *Journal of Computational and Graphical Statistics*, 13(4):907–929, 2004. doi: 10.1198/106186004X12803.
- Jianmin Chen, Xinghao Pan, Rajat Monga, Samy Bengio, and Rafal Jozefowicz. Revisiting distributed synchronous SGD. *arXiv preprint arXiv:1604.00981*, 2016.
- Jean-Marie Cornuet, Jean-Michel Marin, Antonietta Mira, and Christian P. Robert. Adaptive multiple importance sampling. *Scandinavian Journal of Statistics*, 39(4):798–812, 2012. doi: 10.1111/j.1467-9469.2011.00756.x.
- Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020. doi: 10.1073/pnas.1912789117.
- Jeffrey Dean, Greg S. Corrado, Rajat Monga, Kai Chen, Matthieu Devin, Quoc V. Le, Mark Z. Mao, Marc’Aurelio Ranzato, Andrew Senior, Paul Tucker, Ke Yang, and Andrew Y. Ng. Large scale distributed deep networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 25, 2012.

- Pierre Del Moral, Arnaud Doucet, and Ajay Jasra. An adaptive sequential Monte Carlo method for approximate Bayesian computation. *Statistics and Computing*, 22(5):1009–1020, 2012. doi: 10.1007/s11222-011-9271-y.
- Christopher C. Drovandi and Anthony N. Pettitt. Estimation of parameters for macroparasite population evolution using approximate Bayesian computation. *Biometrics*, 67(1):225–233, 2011. doi: 10.1111/j.1541-0420.2010.01410.x.
- Ritabrata Dutta, Marcel Schoengens, Lorenzo Pacchiardi, Avinash Ummadisingu, Nicole Widmer, Pierre Künzli, Jukka-Pekka Onnela, and Antonietta Mira. ABCpy: A high-performance computing perspective to approximate Bayesian computation. *Journal of Statistical Software*, 100(7):1–38, 2021. doi: 10.18637/jss.v100.i07.
- Paul Fearnhead and Dennis Prangle. Constructing summary statistics for approximate Bayesian computation: semi-automatic approximate Bayesian computation. *Journal of the Royal Statistical Society: Series B*, 74(3):419–474, 2012. doi: 10.1111/j.1467-9868.2011.01010.x.
- Rémi Flamary, Nicolas Courty, Alexandre Gramfort, Mokhtar Z. Alaya, Aurélie Boissunon, Stanislas Chambon, Laetitia Chapel, Adrien Corenflos, Kilian Fatras, Nemo Fournier, Léo Gautheron, Nathalie T. H. Gayraud, Hicham Janati, Alain Rakotomamonjy, Ievgen Redko, Antoine Rolet, Antony Schutz, Vivien Seguy, Danica J. Sutherland, Romain Tavenard, Alexander Tong, and Titouan Vayer. POT: Python optimal transport. *Journal of Machine Learning Research*, 22(78):1–8, 2021.
- Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977. doi: 10.1021/j100540a008.
- François Graner and James A. Glazier. Simulation of biological cell sorting using a two-dimensional extended Potts model. *Physical Review Letters*, 69(13):2013–2016, 1992. doi: 10.1103/PhysRevLett.69.2013.
- David Greenberg, Marcel Nonnenmacher, and Jakob Macke. Automatic posterior transformation for likelihood-free inference. In *International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pp. 2404–2414, 2019.
- Peter Hall and C. C. Heyde. *Martingale Limit Theory and Its Application*. Academic Press, New York, 1980.
- Tsuyoshi Hirashima, Elisabeth G. Rens, and Roeland M. H. Merks. Cellular Potts modeling of complex multicellular behaviors in tissue morphogenesis. *Development, Growth & Differentiation*, 59(5):329–339, 2017.
- Emanuel Klinger, David Rickert, and Jan Hasenauer. pyABC: distributed, likelihood-free inference. *Bioinformatics*, 34(20):3591–3593, 2018. doi: 10.1093/bioinformatics/bty361.
- Jochen Kursawe, Ruth E. Baker, and Alexander G. Fletcher. Approximate Bayesian computation reveals the importance of repeated measurements for parameterising cell-based models of growing tissues. *Journal of Theoretical Biology*, 443:66–81, 2018.
- Ben Lambert, Adam L. MacLean, Alexander G. Fletcher, Alexander N. Combes, Melissa H. Little, and Helen M. Byrne. Bayesian inference of agent-based models: a tool for studying kidney branching morphogenesis. *Journal of Mathematical Biology*, 76:1673–1697, 2018.
- Maxime Lenormand, Franck Jabot, and Guillaume Deffuant. Adaptive approximate Bayesian computation for complex models. *Computational Statistics*, 28(6):2777–2796, 2013. doi: 10.1007/s00180-013-0428-3.
- Jarno Lintusaari, Henri Vuollekoski, Antti Kangasrääsiö, Kusti Skytén, Marko Järvenpää, Pekka Marttinen, Michael U. Gutmann, Aki Vehtari, Jukka Corander, and Samuel Kaski. ELFI: Engine for likelihood-free inference. *Journal of Machine Learning Research*, 19(16):1–7, 2018.
- Jan-Matthis Lueckmann, Jan Boelts, David Greenberg, Pedro Gonçalves, and Jakob Macke. Benchmarking simulation-based inference. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 130 of *Proceedings of Machine Learning Research*, pp. 343–351, 2021.

- Jean-Michel Marin, Pierre Pudlo, and Mohammed Sedki. Consistency of adaptive importance sampling and recycling schemes. *Bernoulli*, 25(3):1977–1998, 2019. doi: 10.3150/18-BEJ1042.
- Lawrence M. Murray, Anthony Lee, and Pierre E. Jacob. Parallel resampling in the particle filter. *Journal of Computational and Graphical Statistics*, 25(3):789–805, 2016. doi: 10.1080/10618600.2015.1062015.
- Lawrence M. Murray, Sumeetpal S. Singh, and Anthony Lee. Anytime Monte Carlo. *Data-Centric Engineering*, 2:e7, 2021. doi: 10.1017/dce.2021.6.
- Art Owen and Yi Zhou. Safe and effective importance sampling. *Journal of the American Statistical Association*, 95(449):135–143, 2000. doi: 10.1080/01621459.2000.10473909.
- Brooks Paige, Frank Wood, Arnaud Doucet, and Yee Whye Teh. Asynchronous anytime sequential Monte Carlo. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 27, 2014.
- George Papamakarios and Iain Murray. Fast ϵ -free inference of simulation models with Bayesian conditional density estimation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, 2016.
- G. D. Rayner and H. L. MacGillivray. Numerical maximum likelihood estimation for the g-and-k and generalized g-and-h distributions. *Statistics and Computing*, 12(1):57–75, 2002. doi: 10.1023/A:1013120305780.
- Benjamin Recht, Christopher Ré, Stephen J. Wright, and Feng Niu. Hogwild!: A lock-free approach to parallelizing stochastic gradient descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 24, 2011.
- Yannik Schälte, Emanuel Klinger, Paul Stapor, and Jan Hasenauer. pyABC 0.12.3: high-performance distributed, likelihood-free inference. *Bioinformatics*, 38(13):3354–3355, 2022. doi: 10.1093/bioinformatics/btac321.
- Scott A. Sisson, Yanan Fan, and Mark M. Tanaka. Sequential Monte Carlo without likelihoods. *Proceedings of the National Academy of Sciences*, 104(6):1760–1765, 2007. doi: 10.1073/pnas.0607208104.
- Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. Validating Bayesian inference algorithms with simulation-based calibration. *arXiv preprint arXiv:1804.06788*, 2018.
- Oskar Taubert, Marie Weißer, Daniel Kelle, Markus Bayer, Hilmar Knüttel, Claudia Comito, Achim Streit, and Markus Götz. Massively parallel genetic optimization through asynchronous propagation of populations. In *International Conference on High Performance Computing (ISC High Performance)*, pp. 106–124, 2023. doi: 10.1007/978-3-031-32041-5_6.
- Tina Toni, David Welch, Natalja Strelkowa, Anders Ipsen, and Michael P. H. Stumpf. Approximate Bayesian computation scheme for parameter inference and model selection in dynamical systems. *Journal of the Royal Society Interface*, 6(31):187–202, 2009. doi: 10.1098/rsif.2008.0172.
- Eric Veach and Leonidas J. Guibas. Optimally combining sampling techniques for Monte Carlo rendering. In *Proceedings of SIGGRAPH ’95*, pp. 419–428, 1995. doi: 10.1145/218380.218498.
- Christelle Vergé, Cyrille Dubarry, Pierre Del Moral, and Éric Moulines. On parallel implementation of sequential Monte Carlo methods: The island particle model. *Statistics and Computing*, 25(2):243–260, 2015. doi: 10.1007/s11222-013-9429-x.
- Richard D. Wilkinson. Approximate Bayesian computation (ABC) gives exact results under the assumption of model error. *Statistical Applications in Genetics and Molecular Biology*, 12(2):129–141, 2013. doi: 10.1515/sagmb-2013-0010.

A The central limit theorem and the fidelity ratio

This appendix contains the central limit theorem and its per-run confidence intervals, a bound on how far the tilt by r can move the target, and the decomposition and partial measurement of r on stored runs.

Relation to existing AMIS theory. For a fixed proposal, the r -tilted limit of Theorem 1 is the elementary observation that self-normalized importance sampling against a mismatched denominator converges to a correspondingly tilted target. The present scheme, however, is adaptive at every draw, so the argument runs through a martingale array rather than an i.i.d. one, and the available AMIS results do not cover it: the convergence argument of Cornuet et al. (2012) is heuristic, and Marin et al. (2019) restrict the adaptation so that the weighting stage decouples, which a single-arrival scheme does not do. The prior floor in the *denominator* bounds every weight by $1/\delta$ with no moment or density-ratio condition. This makes the conditional Lindeberg step trivial and lets the method use a local archive proposal at all, for which the usual two-sided ratio bound fails. The departure from the ideal is carried as r inside the statements, then decomposed and partly measured.

Assumption 5(ii), in full. In addition to clause (i), ϵ_∞ and r are deterministic, a (possibly random) limit $\tilde{q}_\infty$ of $\tilde{q}_i$ and $\bar{q}_\infty$ of $\bar{q}_n$ exist with $r = \tilde{q}_\infty/\bar{q}_\infty$, and, almost surely,

$$\begin{aligned} \sqrt{n} (|\epsilon_n - \epsilon_\infty| + \|\bar{q}_n - \bar{q}_\infty\|_\infty) &\rightarrow 0, \\ \frac{1}{\sqrt{n}} \sum_{i \leq n} (|\epsilon_{i-1} - \epsilon_\infty| + \|\tilde{q}_i - \tilde{q}_\infty\|_\infty + \|\bar{q}_{i-1} - \bar{q}_\infty\|_\infty) &\rightarrow 0, \end{aligned} \quad (11)$$

where ϵ_{i-1} and $\bar{q}_{i-1}$ are the bandwidth and the denominator (7) formed from the first $i-1$ draws (both known before draw i ; convention $\bar{q}_0 := \pi$). The two clauses differ in strength: consistency permits r and ϵ_∞ to be random, while the central limit theorem needs them deterministic, because the centered integrand must be a fixed function for the martingale arrays to be adapted. That restriction is vacuous in the case of most interest, $r \equiv 1$. The Lipschitz constant of Assumption 1 is stated per compact sub-interval because it degrades as $\epsilon \downarrow 0$; no deterministic bandwidth floor is needed, since the proofs localize over $[1/j, \epsilon_0]$ and intersect countably many almost-sure events (Appendix B).

Theorem 3 (central limit theorem). *Let Assumptions 1–4 and 5(i)–(ii) hold. Then for every bounded measurable f ,*

$$\sqrt{n}(\hat{\pi}_n(f) - \pi_{\epsilon_\infty}^r(f)) \longrightarrow \sigma_f \mathcal{Z} \quad \text{stably}, \quad \mathcal{Z} \sim \mathcal{N}(0, 1) \text{ independent of } \mathcal{F}_\infty, \quad (12)$$

where the $\mathcal{F}_\infty$ -measurable σ_f^2 is

$$\sigma_f^2 = \frac{1}{Z_r^2} \int_{\Theta} \pi(\theta)^2 (f(\theta) - \pi_{\epsilon_\infty}^r(f))^2 L_{\epsilon_\infty}^{(2)}(\theta) \frac{\tilde{q}_\infty(\theta)}{\bar{q}_\infty(\theta)^2} d\theta, \quad Z_r = \int_{\Theta} \pi L_{\epsilon_\infty} r. \quad (13)$$

Theorem 3 covers variants whose adaptation is frozen after a finite burn-in or damped fast enough for the rate condition (11); whether the data-driven schedule our experiments run meets that condition is measured in §A.1, where the production runs sit at its threshold. At $r \equiv 1$ the variance collapses to $Z_{\epsilon_\infty}^{-2} \int \pi^2 (f - \pi_{\epsilon_\infty} f)^2 L_{\epsilon_\infty}^{(2)}/\bar{q}_\infty$, the familiar self-normalized form. Because $L_\epsilon^{(2)} \geq L_\epsilon^2$ by Jensen, (13) is never smaller than the variance of the corresponding *noise-free* importance sampler that could evaluate L_{ϵ_∞} exactly; the gap $L_\epsilon^{(2)} - L_\epsilon^2 = \text{Var}(K_\epsilon(\rho) \mid \theta)$ is the price of running one simulation per proposed parameter. For the Gaussian kernel $K_\epsilon(\rho)^2 = K_{\epsilon/\sqrt{2}}(\rho)$, so $L_\epsilon^{(2)} = L_{\epsilon/\sqrt{2}}$ and that price is computable from the same likelihood at a tightened bandwidth. Convergence is stable rather than merely distributional because $\tilde{q}_\infty$, and hence σ_f^2 , may be random; it reduces to an ordinary $\mathcal{N}(0, \sigma_f^2)$ limit when the limiting proposal is deterministic. Stability also makes the theorem usable, because it survives division by a consistent variance estimate:

Corollary 4 (per-run confidence intervals). *Under the assumptions of Theorem 3, and provided $\sigma_f^2 > 0$, with $\hat{\sigma}_n^2 := n \sum_i W_{i,n}^2 (f(\theta_i) - \hat{\pi}_n(f))^2 / (\sum_i W_{i,n})^2$ we have $\hat{\sigma}_n^2 \rightarrow \sigma_f^2$ a.s. and $\sqrt{n}(\hat{\pi}_n(f) - \pi_{\epsilon_\infty}^r(f))/\hat{\sigma}_n \Rightarrow \mathcal{N}(0, 1)$. The interval $\hat{\pi}_n(f) \pm z_{1-\alpha/2} \hat{\sigma}_n/\sqrt{n}$ therefore has asymptotic level $1 - \alpha$ and is computable from the stored weights alone, at no extra simulation cost.*

Proposition 5 (how far the tilt can move the target). *Let $A \subseteq \Theta$ and suppose $|r - 1| \leq \eta_A$ on A and $|r - 1| \leq 1$ off it. Then with $\zeta := \eta_A + \pi_{\epsilon_\infty}(A^c)$,*

$$\zeta < 1 \implies \|\pi_{\epsilon_\infty}^r - \pi_{\epsilon_\infty}\|_{\text{TV}} \leq \frac{\zeta}{1 - \zeta}. \quad (14)$$

More sharply, ζ may be replaced by $\zeta_0 = \pi_{\epsilon_\infty}(|r - 1|)$. Estimating ζ_0 from a stored run requires untilting: the normalized weights $\hat{w}_i$ target π^r , not π_{ϵ_∞} , so $\sum_i \hat{w}_i |r(\theta_i) - 1|$ estimates $\pi^r(|r - 1|)$; weights proportional to $W_{i,n}/r(\theta_i)$ target π_{ϵ_∞} and give $\hat{\zeta}_0$.

A.1 What r is for this implementation

We decompose r by inserting intermediate mixtures between the density actually sampled from and the function used in the denominator, changing one implementation feature at a time. Let $\bar{q}^{(1)} = \frac{1}{n} \sum_{i \leq n} q_i$ use the *nominal* archive proposals (the truncated mixtures the propagator intends, with exact normalizing constants and without the rejection cap or the underflow redraws); let $\bar{q}^{(2)}$ replace those constants by the implementation’s product-of-marginals values; and let $\bar{q}^{(3)}$ be the m -point draw-proportional version of $\bar{q}^{(2)}$ carrying the observed bootstrap share ν_n . Assume each of the four displayed ratios has a uniform limit, finite and bounded away from zero. Fidelity alone does not give this, since (9) constrains only their product. Writing $\bar{q}$ for the m -snapshot denominator (7),

$$r = \underbrace{\lim \frac{\bar{q}^*}{\bar{q}^{(1)}}}_{\text{(d) cap, redraws, asynchrony}} \cdot \underbrace{\lim \frac{\bar{q}^{(1)}}{\bar{q}^{(2)}}}_{\text{(c) normalizer}} \cdot \underbrace{\lim \frac{\bar{q}^{(2)}}{\bar{q}^{(3)}}}_{\text{(b) quadrature}} \cdot \underbrace{\lim \frac{\bar{q}^{(3)}}{\bar{q}}}_{\text{(a) floor}}.$$

The product telescopes for any positive intermediates, so the content is in the choice of intermediates: a different insertion order attributes the total differently, and the factors are a decomposition of one quantity rather than four independent errors.

(a) *The prior floor.* When genuine bootstrap draws carry less than a δ share of the history, $w_n = \delta > \nu_n$ and the denominator holds more prior mass than the sample does. With $u = q_\infty/\pi$ the limiting proposal-to-prior ratio, this contributes $r_{\text{floor}} = (\nu_\infty + (1 - \nu_\infty)u) / (\delta + (1 - \delta)u)$, so $r_{\text{floor}} - 1 = (\delta - \nu_\infty)(u - 1) / (\delta + (1 - \delta)u)$: it is exactly 1 when $\nu_\infty \geq \delta$, below 1 wherever the proposal is thinner than the prior, and $r_{\text{floor}} \rightarrow 1$ as $\delta \rightarrow 0$, i.e. as m grows. This is the one contribution available in closed form.

(b) *The snapshot quadrature.* The m -point mixture stands in for the n -point one, with error at most V_n/m where $V_n = \sum_{i < n} \|q_{i+1} - q_i\|_\infty$ is the total variation of the *reconstructed* proposal path; this controls the third factor above, not the fidelity of the whole. Fixing m therefore trades cost against fidelity; see the remark below.

(c) *The truncation constant.* Each mixture component is normalized by the product of its per-axis in-box masses, which is exact for a diagonal perturbation covariance and increasingly accurate as a component’s mass sits away from the faces of the box relative to its own scale, but an approximation otherwise.

(d) *Capped rejection, underflow redraws, and (asynchronously) the filtration gap.* A parent’s perturbation is rejection-sampled into the box for at most 992 attempts, so component j is realized with probability $W_j[1 - (1 - Z_j)^{992}]$ and the residual is emitted as a prior draw; and a candidate whose proposal-time denominator falls below 10^{-12} is redrawn up to five times, the last one being emitted regardless. The cap changes the law relative to unlimited rejection whenever $Z_j < 1$, through the residual probability $(1 - Z_j)^{992}$; $Z_j \approx 10^{-3}$ is the scale below which that residual becomes appreciable, not a threshold above which it vanishes. Under asynchronous execution this factor additionally carries the gap between the emitted proposal density and the conditional law of Assumption 4, which is a property of the schedule.

Empirical estimates of the prior-floor and snapshot-quadrature contributions. These two contributions can be estimated rather than bounded because $\hat{\pi}_n$ is a post-hoc pass. Recomputing (7) at a much larger m and at the observed prior share gives a reference against which they are evaluated at every particle. The truncation constant and factor (d) are bounded rather than measured. On the five Cellular Potts production replicates (about 12,900 asynchronous evaluations each across 48 workers, two correlated dimensions, so that (c) is live; Appendix E) both factors are active: the bootstrap draws make up 1.1% of each history against a floor of $\delta = 2.4\%$, so (a) contributes $\hat{\zeta} = 0.011$ and (b) 0.024–0.028, together 0.035–0.039 with $r \in [0.24, 1.41]$. The posterior estimate lies 0.2–0.5% in total variation from the reference at $m = 400$ and the observed prior share, with posterior means within 0.012 standard deviations, marginal

widths within 1% (narrower in nine of ten, the tail-suppressing direction that (a) predicts), and the effective sample size unchanged. On a one-dimensional Gaussian-mean history where the exact n -point mixture is computable, (a) and (b) together give $\hat{\zeta} = 0.042$ and a total-variation distance of 0.26% from the reference that removes (a) and (b); there too the estimate is 0.5% narrower. Of contribution (d), the two events a stored history records, a rejection fallback and an exhausted redraw, occurred neither in the 12,000 draws of that history nor in the 63,699 archive-phase draws of the five Cellular Potts replicates; a redraw that succeeded on a later attempt leaves no trace, so this is a necessary condition for the event, not a bound on its probability. The bound (14) is loose against these direct measurements by an order of magnitude on both (3.6–4.0% against 0.2–0.5% on Cellular Potts, 4.4% against 0.26% on the one-dimensional history), as sup-norm-free bounds valid for every f generally are.

Do the reported runs meet the stabilization conditions? The proposal sequence is reconstructible from a stored history, so we assess Assumption 5(ii) empirically. Rebuilding q_τ along the five Cellular Potts production replicates and comparing against each run’s terminal proposal separates two rates that behave differently. The *per-step drift* $\|q_{\tau+1} - q_\tau\|_\infty$ decays as $\tau^{-0.7}$ to $\tau^{-0.95}$, and the archive’s membership changes only 825–874 times in about 12,900 draws, tracking $k \log n$ at a ratio between 0.54 and 0.92 across the whole run. This is the rate a record-statistics heuristic predicts, since the k -th best of n draws improves with probability $\approx k/n$. The proposal path’s total variation is therefore $O(\log n)$, and the snapshot clause is met. The *distance to the limit* is the harder condition. Fitting $\|q_\tau - q_\infty\|_\infty \sim \tau^{-b}$ gives $b = 0.50$ – 0.71 per replicate over the full range (0.59 pooled) and $b = 0.25$ – 0.72 (0.48 pooled) when the fit is restricted to the first half, away from the end where using the terminal proposal as a stand-in for q_∞ biases b upward. Assumption 5(ii) needs $b > \frac{1}{2}$, at which the aggregate $n^{-1/2} \sum_{i \leq n} \|q_i - q_\infty\|_\infty$ vanishes like $n^{1/2-b}$: the production runs sit at that threshold.

The measurement spans one benchmark over 1.3 decades of n per replicate, against a stand-in limit, so it is a diagnostic rather than an asymptotic statement. Theorem 3 is accordingly stated for the damped and frozen variants; Theorem 1, needing no rate, is unaffected. The exponent depends on the inference problem, not on the schedule. A tightening bandwidth does not move it: running a two-dimensional Gaussian-mean benchmark serially under the same propagator, once with the data-driven schedule and once with the bandwidth pinned at the tolerance the first run reached a third of the way in, moves b by less than 0.07, and both runs satisfy $b > \frac{1}{2}$ at $b = 0.61$ – 0.76 .

Identifiability does. Re-running the serial benchmark with its two parameters confounded (observing only $\theta_1 + \theta_2$, so the target is a ridge rather than a point) collapses the exponent to $b = 0.00$ – 0.29 : on a ridge the top- k archive slides along the flat direction indefinitely, so the proposal never settles. Assumption 5(ii) is thus a condition on the inference problem as much as on the sampler. It holds with a margin for a point-identified target under the same top- k rule and bandwidth schedule and fails on a ridge; the Cellular Potts target, which resolves two directions but the second less sharply than the first (Appendix E), lies between the two.

Fixed versus growing m . Contribution (b) is where the estimator’s cost meets its fidelity. With m fixed, the m -point mixture tracks the n -point one only if the proposal path settles, the error being V_n/m . Letting $m = m_n \rightarrow \infty$ with $V_n/m_n \rightarrow 0$ removes *this* contribution to r (which does not by itself give $r \equiv 1$, since the other three factors remain), and the proofs go through for any $m_n = o(n/\log n)$ *provided the floor δ is held fixed rather than tied to m* . The implementation ties them, $\delta = 0.5/(m+1)$, so growing m there would send $\delta \rightarrow 0$, and with it the uniform weight bound $1/\delta$ of Lemma 6 that makes the Lindeberg step trivial and fixes the constants in the law of large numbers; the rate conditions would have to be redone. The post-hoc pass can decouple the two (Appendix B, Remark 2). The retroactive pass then costs $\mathcal{O}(n m_n k)$ instead of $\mathcal{O}(nk)$, and the literal cumulative mixture is the endpoint $m_n = n$ at $\mathcal{O}(n^2 k)$. The measurements above locate the practical knee: on the Gaussian-mean history the posterior estimate is within 3×10^{-4} in total variation of the exact n -point mixture by $m = 50$, so a $2.5\times$ more expensive post-hoc pass removes factor (b) almost entirely. The paper reports the $m = 21$ end; §3.5 gives its cost.

B Proofs

B.1 Setting

We work with a filtration $(\mathcal{F}_i)$ for which, writing $\mathbb{E}_{i-1}[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_{i-1}]$, the pair $\xi_i = (\theta_i, \rho_i)$ is $\mathcal{F}_i$ -measurable and the two conditional laws below hold. Particles are indexed by *proposal* time, not arrival time. This indexing is why the martingale structure survives asynchronous execution: a worker forms its proposal from the history it has received at that moment, which is part of $\mathcal{F}_{i-1}$, so the conditional density $\tilde{q}_i$ of the emitted candidate is $\mathcal{F}_{i-1}$ -measurable and

$$\theta_i \mid \mathcal{F}_{i-1} \sim \tilde{q}_i, \quad \rho_i \mid \mathcal{F}_{i-1} \vee \sigma(\theta_i) \sim p(\cdot \mid \theta_i). \quad (15)$$

The proofs index by *proposal* time, whereas the implementation's post-hoc estimator consumes a rank's *arrival*-ordered log and reconstructs the q_{τ_s} from prefixes of it; the two agree for a sequential run and need not otherwise, and the discrepancy is one more contributor to r . Under sequential execution $\mathcal{F}_i = \sigma(\xi_1, \dots, \xi_i)$ serves and (15) holds by construction. Under asynchronous execution a discrepancy ρ_i may still be in flight when candidate $i+1$ is proposed, so the σ -field of everything realized by that moment does not contain ξ_i . One must take a filtration large enough to contain the in-flight outcomes, and Assumption 4 asserts that such a filtration exists *with $\tilde{q}_i$ still the conditional law of the candidate under it*. That is the substantive clause. Conditioning on the larger field need not leave the candidate's law unchanged when completion times steer which rank proposes next. In that case $\tilde{q}_i$ is the enlarged-field conditional law rather than the propagator's emitted density, and (9) is a statement about that object: the completion-time selection of §8, seen from the theory side. Concurrency also affects the reconstruction of the q_{τ_s} from a single rank's arrival-ordered log, which enters only through $\|\bar{q}_n - \bar{q}_\infty\|_\infty$ in Assumption 5.

No structural form is imposed on $\tilde{q}_i$: it is whatever the archive rule, the capped rejection, the underflow redraws and the bootstrap phase jointly produce, and it enters the proofs only through $\bar{q}_n^* = \frac{1}{n} \sum_{i \leq n} \tilde{q}_i$ of (8), for which the denominator $\bar{q}_n$ of (7) is an m -component stand-in. Let $\mathcal{Q}$ be the set of positive functions of the form (7) whose components satisfy Assumption 2; nothing below requires an element of $\mathcal{Q}$ to integrate to one. Elements of $\mathcal{Q}$ are parameterized by $p_q = (\text{means, component weights, covariances, } \alpha)$ ranging over a compact set $P_q \subset \mathbb{R}^D$, $D = m(kd + k + d(d+1)/2) + m$.

Lemma 6 (bounded weights). *Under Assumptions 2–3, $0 \leq W_{i,n} \leq 1/\delta$ for all $i \leq n$ and all n , surely, whatever the sampler does.*

Proof. All components of (7) are non-negative, so $\bar{q}_n \geq \delta\pi$ pointwise on Θ . Hence $W_{i,n} = \pi(\theta_i)K_{\epsilon_n}(\rho_i)/\bar{q}_n(\theta_i) \leq K_{\epsilon_n}(\rho_i)/\delta \leq 1/\delta$, the last step by $K_\epsilon \leq 1$ (Assumption 1). The prior cancels exactly, so the bound needs neither a density-ratio condition nor a moment condition. $\square$

Lemma 7 (uniform SLLN for bounded martingale differences). *Let $P \subset \mathbb{R}^D$ be compact and $\{\psi_p\}_{p \in P}$ measurable functions on a measurable space $\mathcal{X}$ with $\sup_p \|\psi_p\|_\infty \leq B$ and $|\psi_p(x) - \psi_{p'}(x)| \leq \Lambda \|p - p'\|$ for every $x \in \mathcal{X}$. Let (ξ_i) be $\mathcal{X}$ -valued with ξ_i $\mathcal{F}_i$ -measurable. Then*

$$\sup_{p \in P} \left| \frac{1}{n} \sum_{i \leq n} (\psi_p(\xi_i) - \mathbb{E}_{i-1} \psi_p(\xi_i)) \right| \longrightarrow 0 \quad a.s.$$

Proof. Fix p . The variables $D_i^p := \psi_p(\xi_i) - \mathbb{E}_{i-1} \psi_p(\xi_i)$ form a martingale difference sequence with $|D_i^p| \leq 2B$, so by the Azuma–Hoeffding inequality $\mathbb{P}(|\sum_{i \leq n} D_i^p| \geq nt) \leq 2 \exp(-nt^2/8B^2)$ for every $t > 0$. Fix $\eta > 0$ and let P_η be a finite η -net of P , of cardinality at most $C_P \eta^{-D}$. A union bound gives $\mathbb{P}(\max_{p \in P_\eta} |\sum_{i \leq n} D_i^p| \geq nt) \leq 2C_P \eta^{-D} \exp(-nt^2/8B^2)$, which is summable in n (the net is fixed, so its cardinality does not grow with n); by Borel–Cantelli, $\limsup_n \max_{p \in P_\eta} |\frac{1}{n} \sum_{i \leq n} D_i^p| \leq t$ a.s. For arbitrary $p \in P$ pick $p' \in P_\eta$ with $\|p - p'\| \leq \eta$; then $|D_i^p - D_i^{p'}| \leq 2\Lambda\eta$ by the Lipschitz hypothesis (which also bounds the conditional expectations), so $\limsup_n \sup_{p \in P} |\frac{1}{n} \sum_{i \leq n} D_i^p| \leq t + 2\Lambda\eta$ a.s. Intersecting the almost-sure events over the countable family $t = 1/r$, $\eta = 1/j$ with $r, j \in \mathbb{N}$, gives the claim. $\square$

Lemma 8 (conditional-mean identity). *Let $\phi : \Theta \rightarrow \mathbb{R}$ be bounded measurable, let $\ell \in \{1, 2\}$ (an exponent, not the fidelity ratio r), and let $\epsilon \in [a, \epsilon_0]$ for some $a > 0$ and the positive function $q \geq \delta\pi$ be either deterministic or $\mathcal{F}_{i-1}$ -measurable. Then under (15), writing $L_\epsilon^{(1)} := L_\epsilon$,*

$$\mathbb{E}_{i-1} \left[\frac{\phi(\theta_i) K_\epsilon(\rho_i)^\ell}{q(\theta_i)^\ell} \right] = \int_{\Theta} \phi L_\epsilon^{(\ell)} \frac{\tilde{q}_i}{q^\ell} d\theta.$$

Proof. Condition on $\mathcal{F}_{i-1}$, given which ϵ and q are constants, and integrate first over $\rho_i \sim p(\cdot \mid \theta_i)$, which gives the factor $L_\epsilon^{(\ell)}(\theta_i) = \mathbb{E}[K_\epsilon(\rho)^\ell \mid \theta_i]$, then over $\theta_i \sim \tilde{q}_i$. The $\mathcal{F}_{i-1}$ -measurable case is what Steps 2 and 3 of the proof of Theorem 3 need; an $\mathcal{F}_\infty$ -measurable q , such as the limit $\bar{q}_\infty$, would *not* be admissible here, which is why that proof centers on predictable proxies rather than on the limit. Fubini applies because the integrand is bounded by $\|\phi\|_\infty / (\delta\pi_-)^\ell$. The case $\ell = 1$, $\phi = \pi h$ is the one used for first moments; $\ell = 2$, $\phi = \pi^2 h^2$ gives the conditional second moments of Step 3(b) and Corollary 4. $\square$

The class used below. Take $\mathcal{X} = \Theta \times [0, \infty)$, $\xi_i = (\theta_i, \rho_i)$, and, for a fixed bounded measurable h ,

$$\psi_{(\epsilon, p_q)}(\theta, \rho) := \frac{\pi(\theta) h(\theta) K_\epsilon(\rho)}{q_{p_q}(\theta)}, \quad (\epsilon, p_q) \in P_a := [a, \epsilon_0] \times P_q, \quad a \in (0, \epsilon_0]. \quad (16)$$

Each P_a is compact and deterministic, which matters: Lemma 7 cannot be applied over a parameter set that depends on the run, and $[\epsilon_\infty, \epsilon_0]$ would be such a set when ϵ_∞ is random. We therefore apply the lemma on $P_{1/j}$ for every $j \in \mathbb{N}$ with $1/j \leq \epsilon_0$ and intersect the countably many almost-sure events; since $\epsilon_n \downarrow \epsilon_\infty > 0$ a.s., each path has $\epsilon_n \in [1/j, \epsilon_0]$ for all n once $1/j \leq \epsilon_\infty$, so the resulting uniform bound covers $(\epsilon_n, \bar{q}_n)$ on that path. No deterministic bandwidth floor is required. On P_a , $\sup \|\psi\|_\infty \leq \|h\|_\infty / \delta$ by the computation of Lemma 6. For the Lipschitz hypothesis: each truncated component density $\varphi(\theta; \mu_j, \Sigma) / Z(\mu_j, \Sigma)$ has parameter-derivatives bounded uniformly over $\theta \in \Theta$, because Θ is compact, $\sigma_-^2 I \preceq \Sigma \preceq \sigma_+^2 I$ bounds φ and its derivatives, and the truncation mass is bounded below, $Z(\mu_j, \Sigma) \geq c_0 > 0$, uniformly for $\mu_j \in \Theta$. Here it matters that the implementation's constant is the *product of per-axis* in-box masses, not the exact joint mass: each factor is $\Phi((\text{hi}_d - \mu_{jd})/\sigma_d) - \Phi((\text{lo}_d - \mu_{jd})/\sigma_d)$, and since μ_{jd} lies in $[\text{lo}_d, \text{hi}_d]$ that interval contains a half-width $w_d/2$ on one side of μ_{jd} , giving a factor at least $\Phi(w_d/2\sigma_+) - \frac{1}{2} > 0$; the product over coordinates, $\prod_d (\Phi(w_d/2\sigma_+) - \frac{1}{2})$, is therefore bounded below by a constant depending only on the box and σ_+ , and is smooth in (μ_j, Σ) with bounded derivatives there. This is where Assumption 2's box hypothesis is used; compactness of Θ alone would not give it. Hence every $q \in \mathcal{Q}$ (a set of positive functions, not necessarily normalized; Assumption 2) obeys $\delta\pi_- \leq q \leq q_+ < \infty$ and $q \mapsto q(\theta)$ is Lipschitz in p_q uniformly in θ ; so is $1/q$, since q is bounded away from 0. With K_ϵ Lipschitz in ϵ (Assumption 1) and πh bounded, (16) satisfies the hypotheses of Lemma 7. Note that on $\{1/j \leq \epsilon_\infty\}$ we have $(\epsilon_n, \bar{q}_n) \in P_{1/j}$ for every n , and that $\bar{q}_\infty \geq \delta\pi$ by taking uniform limits.

B.2 Proof of Theorem 1

Work on the event $\{1/j \leq \epsilon_\infty\}$ for a fixed j , so that $\hat{p}_n := (\epsilon_n, \bar{q}_n) \in P_{1/j}$ for every n ; the conclusion then follows on the union over j , which has probability one. Fix a bounded measurable h and define, for each fixed $p \in P_{1/j}$, $\Gamma_i(p) := \mathbb{E}_{i-1}[\psi_p(\xi_i)]$ and $\bar{\Gamma}_n(p) := \frac{1}{n} \sum_{i \leq n} \Gamma_i(p)$. On the almost-sure event of Lemma 7 for $P_{1/j}$ the bound holds simultaneously for all $p \in P_{1/j}$, so the function $\bar{\Gamma}_n$ may be evaluated at the random $\hat{p}_n$:

$$\frac{1}{n} \sum_{i \leq n} W_{i,n} h(\theta_i) = \bar{\Gamma}_n(\hat{p}_n) + o(1) \stackrel{\text{Lem. 8}}{=} \int_{\Theta} \pi h L_{\epsilon_n} \frac{\bar{q}_n^*}{\bar{q}_n} d\theta + o(1) \quad \text{a.s.}, \quad (17)$$

using $\frac{1}{n} \sum_{i \leq n} \bar{q}_i = \bar{q}_n^*$. The middle term is the function $\bar{\Gamma}_n(\cdot)$ evaluated at $\hat{p}_n$ and must not be read as $\mathbb{E}_{i-1}[\psi_{\hat{p}_n}(\xi_i)]$: $\hat{p}_n$ depends on draws after i , is not $\mathcal{F}_{i-1}$ -measurable, and that conditional expectation is not what Lemma 7 controls. Split the integral against $\int \pi h L_{\epsilon_\infty}$:

$$\left| \int \pi h L_{\epsilon_n} \frac{\bar{q}_n^*}{\bar{q}_n} - \int \pi h L_{\epsilon_\infty} r \right| \leq \|h\|_\infty \pi_+ |\Theta| \left[R L_K(1/j) |\epsilon_n - \epsilon_\infty| + \left\| \frac{\bar{q}_n^*}{\bar{q}_n} - r \right\|_\infty \right],$$

on the event $\{1/j \leq \epsilon_\infty\}$ of the localization above, using $|L_{\epsilon_n} - L_{\epsilon_\infty}| \leq L_K(1/j)|\epsilon_n - \epsilon_\infty|$ (Assumption 1 and Jensen; every ϵ_n and ϵ_∞ lies in $[1/j, \epsilon_0]$ there) and $L_{\epsilon_\infty} \leq 1$, and where $R < \infty$ bounds $\bar{q}_n^*/\bar{q}_n$ for all large n . That bound needs no structural bound on $\bar{q}_i$, of which we assume none: (9) sends the ratio uniformly to $r \leq r_+$. Both bracketed terms vanish, the first because $\epsilon_n \rightarrow \epsilon_\infty$ and the second by (9). Hence $\frac{1}{n} \sum_i W_{i,n} h(\theta_i) \rightarrow \int \pi h L_{\epsilon_\infty} r$ a.s. Applying this with $h = f$ and with $h \equiv 1$ and dividing, which is legitimate because $Z_r = \int \pi L_{\epsilon_\infty} r \geq r_- Z_{\epsilon_\infty} > 0$, gives $\hat{\pi}_n(f) \rightarrow \pi_{\epsilon_\infty}^r(f)$ a.s. Corollary 2 is the case $r \equiv 1$. $\square$

Remark 1. Consistency used Assumption 5(i) only through (9) and $\epsilon_n \rightarrow \epsilon_\infty$, and nothing at all about the form of $\bar{q}_i$. The theorem is therefore a statement about the algorithm as implemented, not about an idealized variant of it: any sampler whose draw mixture $\bar{q}_n$ tracks, in the ratio sense of (9), is covered, the archive rule, the capped rejection and the underflow redraws included.

B.3 Proof of Theorem 3 and Corollary 4

Let $g := f - \pi_{\epsilon_\infty}^r(f)$, a fixed bounded function satisfying $\int \pi g L_{\epsilon_\infty} r = 0$ (Assumption 5(ii) makes ϵ_∞ and r , hence $\pi_{\epsilon_\infty}^r(f)$, deterministic). Since $\sum_i W_{i,n} g(\theta_i) = (\sum_i W_{i,n})(\hat{\pi}_n(f) - \pi_{\epsilon_\infty}^r(f))$ identically and $\frac{1}{n} \sum_i W_{i,n} \rightarrow Z_r > 0$ a.s. (Theorem 1 with $h \equiv 1$), by Slutsky's theorem (which preserves stable convergence) it suffices to prove

$$S_n := \frac{1}{\sqrt{n}} \sum_{i \leq n} \frac{\pi(\theta_i) g(\theta_i) K_{\epsilon_n}(\rho_i)}{\bar{q}_n(\theta_i)} \rightarrow \mathcal{N}(0, v^2) \text{ stably}, \quad v^2 := \int_{\Theta} \pi^2 g^2 L_{\epsilon_\infty}^{(2)} \frac{\bar{q}_\infty}{\bar{q}_\infty^2} d\theta, \quad (18)$$

after which $\sigma_f^2 = v^2/Z_r^2$, which is (13).

Step 1 (replacing the parameters by predictable proxies). $\bar{q}_\infty$ and ϵ_∞ are in general only $\mathcal{F}_\infty$ -measurable, so an array built from them is *not* adapted and Lemma 8 does not apply to it. We therefore center on the predictable pair instead: let $\bar{q}_{i-1} \in \mathcal{Q}$ be the denominator (7) formed from the first $i-1$ draws and ϵ_{i-1} the bandwidth in force at the i -th proposal, both $\mathcal{F}_{i-1}$ -measurable, and put

$$V_i := \frac{\pi(\theta_i) g(\theta_i) K_{\epsilon_{i-1}}(\rho_i)}{\bar{q}_{i-1}(\theta_i)}.$$

Since $\bar{q}_n, \bar{q}_{i-1} \geq \delta\pi_-$ and $K \leq 1$, and $|\epsilon_n - \epsilon_{i-1}| \leq |\epsilon_n - \epsilon_\infty| + |\epsilon_{i-1} - \epsilon_\infty|$ (likewise for $\bar{q}$),

$$\left| S_n - \frac{1}{\sqrt{n}} \sum_{i \leq n} V_i \right| \leq C\sqrt{n}(|\epsilon_n - \epsilon_\infty| + \|\bar{q}_n - \bar{q}_\infty\|_\infty) + \frac{C}{\sqrt{n}} \sum_{i \leq n} (|\epsilon_{i-1} - \epsilon_\infty| + \|\bar{q}_{i-1} - \bar{q}_\infty\|_\infty)$$

with $C = \|g\|_\infty \pi_+ \max\{L_K(\epsilon_\infty)/(\delta\pi_-), (\delta\pi_-)^{-2}\}$. This is a single constant, since Assumption 5(ii) makes ϵ_∞ deterministic, so the central limit theorem needs no localization; both brackets vanish by (11). This step is where the rates are used; the remainder of the proof never touches $\bar{q}_n$ or ϵ_n again.

Step 2 (drift). Lemma 8 with $\ell = 1$, $\phi = \pi g$ applies to V_i because $(\epsilon_{i-1}, \bar{q}_{i-1})$ is $\mathcal{F}_{i-1}$ -measurable and g is a fixed function under the central-limit clause of Assumption 5; with $\int \pi g L_{\epsilon_\infty} r = 0$,

$$c_i := \mathbb{E}_{i-1} V_i = \int \pi g \left(L_{\epsilon_{i-1}} \frac{\bar{q}_i}{\bar{q}_{i-1}} - L_{\epsilon_\infty} r \right) d\theta,$$

$$|c_i| \leq C'(|\epsilon_{i-1} - \epsilon_\infty| + \|\bar{q}_i - \bar{q}_\infty\|_\infty + \|\bar{q}_{i-1} - \bar{q}_\infty\|_\infty),$$

C' depending only on $\|g\|_\infty, \pi_\pm, \delta, |\Theta|, L_K(\epsilon_\infty)$. The second half of (11) is exactly the statement that $n^{-1/2} \sum_{i \leq n}$ of that bound tends to zero, so $n^{-1/2} \sum_{i \leq n} c_i \rightarrow 0$. (Under the sufficient condition that each of the three terms is $O(i^{-1/2-a})$ this holds in each of the cases $a < 1/2$, $a = 1/2$, $a > 1/2$, where the partial sums are $O(n^{1/2-a})$, $O(\log n)$ and $O(1)$ respectively.)

Step 3 (martingale CLT). Let $X_{n,i} := n^{-1/2}(V_i - c_i)$, a martingale difference array with respect to the filtration $(\mathcal{F}_i)$, which does not depend on n and is therefore nested. Two conditions: (a) *Conditional Lindeberg.* $|X_{n,i}| \leq 2\|g\|_\infty \pi_+ / (\delta\pi_- \sqrt{n})$ surely, so for every $\eta > 0$ the truncated sum $\sum_i \mathbb{E}_{i-1}[X_{n,i}^2 \mathbf{1}\{|X_{n,i}| > \eta\}]$ is exactly zero for all n large. The defensive floor makes this step trivial where an unbounded-weight

scheme would need a uniform-integrability argument. (b) *Conditional variance.* Lemma 8 with $\ell = 2$, $\phi = \pi^2 g^2$ gives

$$\mathbb{E}_{i-1} V_i^2 = \int \frac{\pi^2 g^2 L_{\epsilon_{i-1}}^{(2)}}{\bar{q}_{i-1}^2} \tilde{q}_i d\theta = v^2 + R_i,$$

$$v^2 = \int \pi^2 g^2 L_{\epsilon_\infty}^{(2)} \frac{\tilde{q}_\infty}{\bar{q}_\infty^2} d\theta, \quad |R_i| \leq C''(|\epsilon_{i-1} - \epsilon_\infty| + \|\tilde{q}_i - \tilde{q}_\infty\|_\infty + \|\bar{q}_{i-1} - \bar{q}_\infty\|_\infty),$$

where C'' carries $2L_K(\epsilon_\infty)$ for the squared kernel, since $|K_\epsilon^2 - K_{\epsilon'}^2| = |K_\epsilon - K_{\epsilon'}|(K_\epsilon + K_{\epsilon'}) \leq 2L_K(\epsilon_\infty)|\epsilon - \epsilon'|$. The bound on $|R_i|$ has vanishing Cesàro average, a fortiori from (11), so $\frac{1}{n} \sum_{i \leq n} \mathbb{E}_{i-1} V_i^2 \rightarrow v^2$ a.s.; with $\frac{1}{n} \sum_i c_i^2 \rightarrow 0$ from Step 2, $\sum_i \mathbb{E}_{i-1} X_{n,i}^2 = \frac{1}{n} \sum_i (\mathbb{E}_{i-1} V_i^2 - c_i^2) \rightarrow v^2$ a.s. Note that v^2 is permitted to be random here: an $\mathcal{F}_\infty$ -measurable limit of the conditional variances is the setting the martingale CLT below covers, and predictability was needed only inside $\mathbb{E}_{i-1}$. The martingale central limit theorem for arrays with nested σ -fields, in its conditional-variance form (Hall & Heyde, 1980, Cor. 3.1, pp. 58–59), then gives $\sum_i X_{n,i} \rightarrow \mathcal{N}(0, v^2)$ stably; the convergence is stable rather than merely distributional because v^2 is $\mathcal{F}_\infty$ -measurable and possibly random. Combining Steps 1–3 gives (18). $\square$

Proof of Corollary 4. Write $\hat{\sigma}_n^2 = [\frac{1}{n} \sum_i W_{i,n}^2 (f(\theta_i) - \hat{\pi}_n(f))^2] / [\frac{1}{n} \sum_i W_{i,n}^2]$. In the numerator, $(f - \hat{\pi}_n(f))^2 = g^2 + 2g(\pi_{\epsilon_\infty}^r(f) - \hat{\pi}_n(f)) + (\pi_{\epsilon_\infty}^r(f) - \hat{\pi}_n(f))^2$, and $W_{i,n} \leq 1/\delta$ with $\hat{\pi}_n(f) \rightarrow \pi_{\epsilon_\infty}^r(f)$ a.s. (Theorem 1), so the cross terms vanish and it suffices to treat $\frac{1}{n} \sum_i W_{i,n}^2 g(\theta_i)^2$. Apply Lemma 7 on P_a with $a = \epsilon_\infty$ (deterministic here, so no localization is needed) to the class $\psi_{(\epsilon, p_q)}(\theta, \rho) = \pi(\theta)^2 g(\theta)^2 K_\epsilon(\rho)^2 / q_{p_q}(\theta)^2$, which is bounded by $\|g\|_\infty^2 / \delta^2$ and Lipschitz in (ϵ, p_q) by the argument of (16) with $2L_K(\epsilon_\infty)$ in place of L_K ; the substitution of $\hat{p}_n$ used in (17), followed by Lemma 8 with $\ell = 2$, $\phi = \pi^2 g^2$ (a fixed function under Assumption 5(ii)), gives $\frac{1}{n} \sum_i W_{i,n}^2 g^2(\theta_i) = \int \pi^2 g^2 L_{\epsilon_n}^{(2)} (\bar{q}_n^* / \bar{q}_n) \bar{q}_n^{-1} + o(1)$, and $\epsilon_n \rightarrow \epsilon_\infty$, $\|\bar{q}_n^* / \bar{q}_n - r\|_\infty \rightarrow 0$ and $\|\bar{q}_n - \bar{q}_\infty\|_\infty \rightarrow 0$ (the last from (11)) send this to $\int \pi^2 g^2 L_{\epsilon_\infty}^{(2)} r / \bar{q}_\infty = \int \pi^2 g^2 L_{\epsilon_\infty}^{(2)} \tilde{q}_\infty / \bar{q}_\infty^2 = v^2$ a.s., using $r = \tilde{q}_\infty / \bar{q}_\infty$. The denominator tends to Z_r^2 , so $\hat{\sigma}_n^2 \rightarrow v^2 / Z_r^2 = \sigma_f^2$ a.s. Because the convergence in Theorem 3 is stable, the pair $(\sqrt{n}(\hat{\pi}_n(f) - \pi_{\epsilon_\infty}^r(f)), \hat{\sigma}_n)$ converges jointly to $(\sigma_f \mathcal{Z}, \sigma_f)$, because stable convergence is equivalent to joint convergence with every $\mathcal{F}_\infty$ -measurable variable and σ_f is one. With $\sigma_f > 0$ a.s. the continuous mapping theorem then gives $\mathcal{Z} \sim \mathcal{N}(0, 1)$ as the limit of the ratio. Marginal convergence to a mixed normal alone would not suffice. $\square$

Remark 2 (growing snapshot count). Nothing above requires m to be fixed except the compactness and fixed dimension D of P_q , and the fixed floor δ . The implementation ties δ to m as $0.5/(m+1)$, so the two must be decoupled before m is allowed to grow, since Lemma 6’s bound is $1/\delta$. With δ fixed and $m = m_n \rightarrow \infty$, run the proof of Lemma 7 with an n -dependent net: the union bound costs $D_n \log(1/\eta_n)$ with $D_n = \Theta(m_n)$, and the Lipschitz constant over the mixture weights grows at most polynomially in m_n , so choosing $\eta_n \rightarrow 0$ fast enough leaves the Borel–Cantelli step summable for any $m_n = o(n/\log n)$. In that regime the snapshot clause of Assumption 5 follows from $V_n/m_n \rightarrow 0$, with $V_n = \sum_{i \leq n} \|q_{i+1} - q_i\|_\infty$ the total variation of the proposal path, rather than having to be assumed: the draw-proportional m_n -point mixture is then a converging quadrature of the empirical mixture. The retroactive pass costs $\mathcal{O}(n m_n k)$.

B.4 Proof of Proposition 5

Write $\zeta_0 := \pi_{\epsilon_\infty}(|r - 1|)$. Under the hypothesis, $\zeta_0 \leq \eta_A + \pi_{\epsilon_\infty}(A^c) = \zeta$. For bounded f , since $\pi_{\epsilon_\infty}^r(f) = \pi_{\epsilon_\infty}(fr) / \pi_{\epsilon_\infty}(r)$,

$$\pi_{\epsilon_\infty}^r(f) - \pi_{\epsilon_\infty}(f) = \frac{\pi_{\epsilon_\infty}(f(r-1)) - \pi_{\epsilon_\infty}(f) \pi_{\epsilon_\infty}(r-1)}{\pi_{\epsilon_\infty}(r)},$$

whose modulus is at most $2\|f\|_\infty \zeta_0 / (1 - \zeta_0)$ because $\pi_{\epsilon_\infty}(r) \geq 1 - \zeta_0$. Taking the supremum over $\|f\|_\infty \leq 1$ and halving bounds the total variation by $\zeta_0 / (1 - \zeta_0)$; as $x \mapsto x/(1-x)$ increases on $[0, 1)$, this is at most $\zeta / (1 - \zeta)$ whenever $\zeta < 1$, which is (14). For $\zeta \geq 1$ the bound degenerates and only the trivial $\|\cdot\|_{\text{TV}} \leq 1$ survives.

The closed form quoted in §A.1(a) for the prior floor’s contribution follows by evaluating (9) for that contribution alone: with $u = q_\infty/\pi$ and a limiting bootstrap share $\nu_\infty < \delta$,

$$r_{\text{floor}} - 1 = \frac{\nu_\infty + (1 - \nu_\infty)u}{\delta + (1 - \delta)u} - 1 = \frac{(\delta - \nu_\infty)(u - 1)}{\delta + (1 - \delta)u},$$

so on $\{u \geq c\}$, splitting at $u = 1$ and bounding each branch, $|r_{\text{floor}} - 1| \leq (\delta - \nu_\infty) \max\{(1 - \delta)^{-1}, (\delta + (1 - \delta)c)^{-1}\}$, which is the η_A of Proposition 5 for $A = \{u \geq c\}$. It is identically 1 when $\nu_\infty \geq \delta$, and $\rightarrow 1$ as $\delta \rightarrow 0$. $\square$

B.5 Status of each assumption in the implementation

C Implementation Details

Relation to classical ABC-SMC. Table 6 sets the method against the generation-staged ABC-SMC/PMC implementation we compare with (the pyABC baseline), property by property. None of the listed properties is intrinsic to ABC-SMC as a family (smooth kernels and richer proposals exist), but each characterizes pyABC as configured. Because all state is a pure function of the history (last row), there is nothing to synchronize between workers, which fits the method to the asynchronous island model of §5.2.

Carried state and restart. Between calls the running sampler carries only the kept snapshots of §3.5 and the scheduler’s two throttles, the search cadence and the acceptance gate described below. A restart therefore reproduces the estimator over whatever history was evaluated, not the trajectory the run would have taken (kill-and-resume, Appendix D).

Hyperparameters. Table 7 lists every setting of the sampler with its option name in the implementation and the value used in the experiments.

Glyphs of Fig. 1. The toy problem has $k = N = 12$. Row (a) is textbook PMC with a quantile schedule and a hard threshold, weights $\omega_i^{(t+1)} \propto \pi(\theta_i)/q_t(\theta_i)$ normalized over the N kept. Point sizes vary where weights do (a1–a3 by $\omega^{(t)}$, a6 by $\omega^{(t+1)}$, b3 by the archive weights $\tilde{W}_j$); shading in b1 is the discrepancy (light: far); dotted ellipses are Σ_t and Σ_n ; gray contours are the current proposal in a5 and b5 and the stored snapshots in b4 and b6; the dashed ellipse in row (a) is the hard acceptance region; the inset in column 2 is the kernel over ρ , the Gaussian K_{ϵ_n} in (b) against the indicator in (a). The thin box in every panel is the prior box, identical throughout. Bootstrap draws are prior draws with $w^* = 1$ and no stored bandwidth.

Tolerance schedulers. Every scheduler is a pure function of the evaluated history and is *monotone by construction*: the propagator clips any proposed bandwidth above the history-reconstructed running minimum ϵ_{hist} (§3.3), so the effective tolerance can only decrease. Let μ denote the extra-individual margin (default $\mu = k$); it is unrelated to the snapshot count m of §4. The three base rules act on the individuals *within* the current bandwidth, i.e. the hard-threshold set $\{i : \rho_i < \epsilon\}$ (this count drives only the tolerance schedule; the proposal mixture and the posterior estimate use the smooth-kernel weight $K_\epsilon(\rho_i)$, to which every evaluated particle contributes):

- **Quantile.** $\epsilon \leftarrow$ the lower-rank (non-interpolated) p -th percentile of the accepted losses, default $p = 50$; applied only once at least $k + m$ individuals are accepted.
- **Geometric decay.** $\epsilon \leftarrow \epsilon_0 \gamma^e$ after each completed epoch e of $k + \mu$ accepted individuals, default $\gamma = 0.9$.
- **Acceptance rate.** Over a sliding window of the most recent $k + \mu$ individuals, if the acceptance rate exceeds r_{hi} (default 0.3) tighten by a factor 0.9, otherwise hold; the rule never expands ϵ (the low-rate guard $r_{\text{lo}} = 0.1$ simply holds).

Table 5: What the implementation guarantees, what it assumes, and what it violates. “By construction” means the code enforces the property for every run, not that it was checked post hoc.

Assumption	Status	Basis
1 kernel, Lipschitz part	by construction	Gaussian and Epanechnikov are peak-normalized and Lipschitz in ϵ on every $[a, \epsilon_0]$, with $L_K(a) = 2/ea$ and $2/a$. The hard indicator is not, and is excluded from the theory.
1 $Z_{\epsilon_\infty} > 0$	holds for Gaussian	The Gaussian kernel is strictly positive, so $L_{\epsilon_\infty} > 0$ and the limiting target is non-degenerate whatever the discrepancy distribution. For a compactly supported kernel (Epanechnikov) it requires the achievable discrepancies to reach inside the support, which we do not verify.
2 denominator class	by construction	The prior support is a box and the archive lies in it, so component means do; the covariance carries an additive jitter $10^{-9}\text{tr}(\widehat{\Sigma})/d + 10^{-12}\bar{b}^2$ ($\bar{b}^2$ the mean squared box width), whose second term is a run-independent floor, giving $\sigma_-^2 > 0$, and σ_+^2 follows from the archive lying in a bounded box. The truncation <i>normalizer</i> is the product of per-axis in-box masses, so with a correlated covariance $\bar{q}_n$ does not integrate to one. Assumption 2 does not require that: it constrains a positive function, and the discrepancy is carried by r instead (§A.1(c)).
4 adapted filtration	assumed	Automatic for a sequential run. Under asynchrony a discrepancy may be in flight when the next candidate is proposed, so the field of realized outcomes cannot serve; the assumption is that a larger filtration exists under which the candidate’s conditional law is still $\bar{q}_i$. Where completion times steer scheduling, $\bar{q}_i$ is that enlarged-field law rather than the emitted proposal density. Same mechanism as the completion-time selection limitation of §8.
3 prior floor	by construction	The retroactive denominator floors the prior component’s mass at $\delta = 0.5/(m+1)$ (Appendix C), so $\bar{q}_n \geq \delta\pi$ and Lemma 6 bounds every posterior weight by $1/\delta$, independently of what the sampler does.
5(i) fidelity (9)	assumed; two of its four factors measured	The one place the sampler enters for Theorem 1. Its four factors are decomposed in §A.1; two of them, the prior floor and the snapshot quadrature, are estimated from stored histories rather than bounded ($\widehat{\zeta} = 0.035\text{--}0.039$ on the five Cellular Potts production replicates, the posterior estimate 0.2–0.5% in total variation from a large- m reference), while the truncation constant and the rejection/redraw conditioning are bounded only. Since the factors multiply, the measured part is a partial accounting rather than a bound on the whole.
5(ii) stabilization	measured; at the threshold	Not implied by the top- k rule, and measured: on the five Cellular Potts production replicates the per-step drift decays as $\tau^{-0.7}$ to $\tau^{-0.95}$ (archive turnover 825–874 changes in about 12,900 draws, the $O(k \log n)$ record rate), while the distance to the limit decays as τ^{-b} with $b = 0.50\text{--}0.71$ over the full run and 0.25–0.72 away from the end (0.59 and 0.48 pooled), against the $b > \frac{1}{2}$ needed. A serial benchmark passes at $b \approx 0.6\text{--}0.76$ when the target is identified and fails at $b \approx 0.0\text{--}0.29$ when its parameters confound; the condition thus concerns the inference problem as much as the sampler. The central limit theorem is stated for the damped or frozen variants; consistency, needing no rate, is unaffected. The asynchronous replay of the proposal sequence enters here too, since a reporting rank’s arrival-ordered log need not reproduce the proposal each candidate came from.
5 $\epsilon_\infty > 0$ (deterministic, for the CLT only)	not satisfied	A strictly positive <code>min_tol</code> forces $\epsilon_\infty \geq \text{min_tol} > 0$, since $\{\epsilon_n\}$ is non-increasing and clipped from below, which is all Theorem 1 needs. Theorem 3 additionally needs ϵ_∞ deterministic, so that $f - \pi_{\epsilon_\infty}(f)$ is a fixed function; a floor alone does not give that, since a monotone data-driven schedule may settle above it at a run-dependent value. A schedule that reaches its floor in finite time does, and makes the ϵ terms of (11) vanish identically. The runs reported here leave <code>min_tol</code> unset (zero is accepted), so neither part is <i>established</i> , though neither is disproved: a data-driven decreasing schedule may converge to a positive limit, even a deterministic one, but we do not verify it. Note also that the bandwidth a finite budget reaches is not itself an asymptotic ϵ_∞ .
5 rate (11)	assumed	Needed only in Step 1 and Step 2 of the proof of Theorem 3; consistency does not use it.

With a smooth (Gaussian or Epanechnikov) kernel the schedulers run in a *kernel-aware* mode: instead of the hard $\rho < \epsilon$ count, ϵ is chosen by bisection so that the per-step kernel-weighted effective-sample-size retention ratio meets a target (default 0.95), which keeps the bandwidth schedule comparable across kernel choices. That search runs on a rank’s first call, then once per `bisect_interval` ($= k$) of its own calls, and again whenever the rank’s cached history is rebuilt after an out-of-order arrival; asynchronous arrivals cause such

Table 6: A typical generation-staged ABC-SMC/PMC implementation (the pyABC baseline used here) versus the proposed generation-free method. The listed properties characterize pyABC as configured, not ABC-SMC as a family. The running sampler does carry the online snapshot buffer of §3.5; the posterior estimate rebuilds it from the history.

Property	Classical ABC-SMC	This work
Update style	generation, barrier-synchronized	event-driven (single-arrival), no barrier
Archive	explicit, generational	reconstructed, sliding top- k
ABC likelihood	hard threshold	smooth kernel
Importance weight	against q_{t-1} only	balance heuristic over $\mathcal{S}$
Archive state	mutable, carried	none; reconstructed per call
Posterior estimate	final population	pure function of the history

Table 7: Settings of the asynchronous sampler. Option names are those of the implementation; values are those of the production experiments, except where a study sweeps them (Appendix F).

Symbol	Meaning	Option	Value
k	archive size	<code>k</code>	100 (200 in the four-dimensional calibration runs)
ϵ_0	starting bandwidth	<code>tol_init</code>	5 (Gaussian mean), 10 (g-and-k), 5×10^5 (Lotka–Volterra), 0.1 (Cellular Potts; 10 in its earlier parameterization)
K_ϵ	ABC kernel	<code>kernel</code>	Gaussian
s	scale of the perturbation covariance Σ_n	<code>perturbation_scale</code>	0.8
S	kept snapshots	<code>amis_snapshots</code>	20
	snapshot cadence, in calls	<code>amis_interval</code>	k
	schedule family	<code>scheduler_type</code>	acceptance rate (geometric decay on Lotka–Volterra and in the four-dimensional calibration runs)
	ESS retention per search	<code>ess_target</code>	0.95
	largest tightening per search	<code>max_tighten_factor</code>	0.5
	search cadence, in calls per rank	<code>bisect_interval</code>	k
μ	acceptance gate: hold until $k + \mu$	<code>additional_needed_inds</code>	k
	particles lie within ϵ		
δ	prior floor in $\tilde{q}_n$	—	$0.5/(m+1)$

rebuilds and lockstepped ones do not, which is why the barrierized twin’s bandwidth tightens less often than the asynchronous sampler’s at equal call counts (§6.1).

Proposal covariance. The archive A_n is the top- k history members by smallest discrepancy. Component weights are formed in log-space as $\tilde{W}_j \propto w_j K_{\epsilon_n}(\rho_j)$ and normalized (a uniform fallback is used if all smooth weights underflow). The perturbation covariance is $\Sigma_n = s[\widehat{\text{Cov}}_{\tilde{W}}(A_n) + \lambda_n I]$ with s the `perturbation_scale` and a two-part jitter $\lambda_n = 10^{-9} \text{tr} \widehat{\text{Cov}}_{\tilde{W}}(A_n)/d + 10^{-12} \bar{b}^2$, where $\bar{b}^2$ is the mean squared prior-box width; it is Cholesky-factored once per call. The first term is scale-aware, so a tightly converged archive (marginal standard deviation $\sim 10^{-4}$) is not swamped by the floor as a fixed 10^{-6} would swamp it; the second is a run-independent absolute floor that keeps the matrix factorable for a degenerate archive and supplies the uniform lower spectral bound Assumption 2 needs (Table 5). Each mixture component is truncated to the prior box, with its log normalizing mass computed as the product of the per-axis in-box masses. That is exact for a diagonal covariance, in which case the proposal integrates to one on the support; for a correlated covariance it is an approximation, so the evaluated proposal is a positive function that need not integrate to one. This is why §4 states its denominator assumption for positive functions and assigns the discrepancy to factor (c) of r (§A.1).

Matched pyABC acceptor. For pyABC, pyABC’s `UniformAcceptor` is replaced, for smooth kernels, by a probabilistic-rejection acceptor that accepts a candidate of discrepancy ρ with probability $K_\epsilon(\rho)$ (peak-

normalized so $K_\epsilon(0) = 1$), using the *same* kernel K_ϵ as the asynchronous sampler; the hard kernel falls back to **UniformAcceptor**. The rejection step of each replicate draws from an independently seeded generator. This is the sense in which the kernel and acceptor are matched across methods (§5.2).

Bounded-memory posterior estimator. The retroactive AMIS estimator (6) evaluates the snapshot denominator $\bar{q}_n$, a draw-proportional mixture of $m \leq S+1$ history-reconstructed proposals (21 at the $S = 20$ of every experiment) with a defensive prior component (mass floored at $\delta = 0.5/(m+1)$), at each of the n evaluated particles. The total cost is $\mathcal{O}(nmk) = \mathcal{O}(nk)$ with m fixed. The snapshot indices are spread over the n_{arch} archive-phase calls (indices 0, step, 2step, $\dots$ with $\text{step} = \max(1, \lfloor n_{\text{arch}}/S \rfloor)$), plus the most recent archive-phase proposal if it is not already among them; hence $m \leq S+1$ rather than S , and $m = 2$ when $S = 0$. Each is rebuilt from the history prefix the call saw, so the pass never reads the online ring buffer of §3.5. For near-instantaneous simulators the history reaches $\mathcal{O}(10^7)$ particles, so the pass is accumulated in chunks of $2^{16} = 65,536$ history points: peak memory is $\mathcal{O}(\text{chunk} \cdot m \cdot k)$ rather than $\mathcal{O}(n)$. Each particle’s denominator is evaluated independently within a single chunk (there is no log-sum-exp reduction *across* chunk boundaries), so the result is bit-identical to the unchunked computation. Benchmarked at $k = 100$, $m = 20$ over n from 2.5×10^4 to 10^6 , the pass costs 49.3 s per 10^6 particles on a single core and is linear in n , so the 3×10^7 -particle Gaussian history extrapolates to ≈ 25 minutes; peak memory is flat in n , as the chunking predicts. The pass runs after the timed budget and is excluded from the throughput measurements; for a near-instantaneous simulator it is comparable to the timed budget itself, for cost-bearing simulators negligible against the simulation time.

D Additional Experimental Protocol

Effective simulating ranks. A “ W -worker” point allocates W physical cores to both samplers. Every rank of the asynchronous sampler evaluates; pyABC’s MPI sampler is root-driven, so rank 0 dispatches without simulating and pyABC has $W-1$ simulating ranks. We report the allocated W for both, which flatters the asynchronous sampler by $1/W$ (2.1% at $W=48$, 0.3% at 384) and is not corrected.

Seeding. Each experiment derives its replicate seeds deterministically from a single base seed via `make_seeds`: a `random.Random(base)` stream emits unique non-negative 31-bit integers, one per replicate, portable across platforms and independent of any global RNG state. Derived seeds (observed-data generation, the pyABC rejection step) use a structured BLAKE2b hash (`stable_seed`) of their inputs, so identical inputs reproduce identical seeds across fresh and resumed (`-extend`) runs.

Crash recovery (kill-and-resume). On the Gaussian-mean benchmark (4 workers, a 1.2×10^5 -simulation budget) a run hard-killed partway through inference and relaunched with an identical command reloads the periodic population checkpoint, completes the same total simulation budget with no duplicated records, and recovers the analytic posterior mean (absolute error 7.6×10^{-3} , against 7.5×10^{-4} for an uninterrupted run). It is not a bit-identical replay (the two posterior means differ by 6.9×10^{-3}), because the candidate generator is re-seeded rather than checkpointed, MPI arrival order is not deterministic, and the online AMIS snapshot buffer is rebuilt empty on restart. The post-restart proposal sequence therefore follows a different, equally valid adaptation trajectory. This is the distinction §3.1 draws: recovery yields a valid posterior over the actual evaluated history, not the identical one.

Cluster and wall-clock control. Experiments run on JUWELS (Forschungszentrum Jülich) under ParaStation MPI, 48 cores per node, scheduled with SLURM. All wall-clock-budgeted methods stop at a fixed wall-clock budget enforced uniformly by a monotonic-clock deadline with first-rank-hit semantics: once any rank trips the deadline it serializes its current state into the records and returns, so a partial final generation is recorded rather than discarded (the simulation-limited studies marked † in Table 8, the SBC and kill-and-resume runs, instead stop at a fixed per-trial simulation count). Posterior estimation and plotting run off the timed inference path, so they do not count against the budget. Matched-budget tolerances (§5.3) count only records of completed simulations; pyABC also writes one record per accepted particle, and mixing the two would flatter it.

Table 8: Resolved experiment configurations. Workers, per-run wall-clock budget, replicates (trials for SBC), archive size k , and methods (A: asynchronous; S: pyABC (matched kernel); R: rejection ABC; T: barrierized twin). The twin row covers the straggler, heterogeneity and Cellular Potts 50³ studies at 16, 48 and 48–384 workers. [†] Simulation-limited: the budget is per-trial (SBC) or per-run (kill-and-resume) *simulations*, not wall-clock seconds; the twin, and the asynchronous re-run matched to it, stop at a fixed simulation count per level (§5.2, Table 9). [‡] Also run with AMIS on/off, a drain-after-deadline variant, and a steeper coupling regime (base delay 0.5 s, cap 8 s). [§] The Cellular Potts rows of these studies ran the benchmark’s earlier configuration (Appendix E).

Experiment	Workers	Budget (s)	Reps	k	Methods
Gaussian mean	48	300	5	100	A,S,R
g-and-k	48	600	5	100	A,S,R
Lotka–Volterra	48	600	5	100	A,S
Cellular Potts 50 ³	48	3600	5	100	A,S,R
Cellular Potts 80 ³	48	3600	3	100	A,S,R
Straggler	16	300	5	100	A,S
Runtime heterogeneity	48	60	5	100	A,S
Parameter-coupled runtime [‡]	48	60	5	100	A,S
Strong scaling (LV)	1–288	180	5	100/1000	A,S
Strong scaling (CPM) [§]	1–384	1800	3–5	100	A,S
Barrierized twin [§]	16 / 48 / 48–384	sims [†]	5	100	T,A
Sensitivity	48	300	5	50/200	A
Ablation	48	300	5	100	A
SBC (Gaussian, 1-D)	48	20k [†]	1000	100	A,S
SBC (bimodal)	48	20k [†]	1000	100	A,S
SBC (g-and-k, 4-D)	48	40k [†]	1000	200	A,S
SBC (g-and-k, 2×)	48	80k [†]	500	200	A,S
Kill-and-resume	4	120k [†]	1	100	A

Benchmark configurations. Table 8 lists the resolved settings (k is the archive/population size; the budget is per method run). The strong-scaling studies sweep the worker count from 1 to 288 (Lotka–Volterra; sub-node 1/4/16 and fully-packed multiples of 48 up to six nodes) and to 384 (Cellular Potts; up to eight nodes), with archive size $k \in \{100, 1000\}$; all other studies fix the workers shown.

Benchmark-specific notes. Each benchmark fixes one observed dataset (generated from seed 42) and infers parameters under uniform priors; the discrepancy is a distance between simulated and observed summary statistics. *Gaussian mean*: $n_{\text{obs}} = 100$ observations $\sim \mathcal{N}(0, 1)$, prior $\mu \sim \mathcal{U}[-5, 5]$, summary the sample mean, discrepancy $|\bar{x}_{\text{sim}} - \bar{x}_{\text{obs}}|$; the closed-form Gaussian posterior is the calibration and recovery reference. *g-and-k*: the quantile-defined g-and-k distribution ($c = 0.8$), $n_{\text{obs}} = 1000$, true $(A, B, g, k) = (3, 1, 2, 0.5)$, priors $A \sim \mathcal{U}[0, 6]$, $B \sim \mathcal{U}[0.1, 4]$, $g \sim \mathcal{U}[0, 5]$, $k \sim \mathcal{U}[0, 1]$; summary the seven octiles (the 0.125, ..., 0.875 quantiles), discrepancy Euclidean. *Lotka–Volterra*: a Gillespie stochastic simulation (Gillespie, 1977) of the four-reaction predator–prey system from $(X_0, Y_0) = (50, 100)$ over $T = 30$, true rates $\theta = (0.5, 0.025, 0.025, 0.5)$, priors $\theta_1, \theta_4 \sim \mathcal{U}[0.05, 2]$ and $\theta_2, \theta_3 \sim \mathcal{U}[0.001, 0.2]$; summary the six statistics $(\bar{X}, \text{sd } X, \bar{Y}, \text{sd } Y, \log(1 + \text{var } X/\bar{X}), \log(1 + \text{var } Y/\bar{Y}))$, discrepancy Euclidean. *Cellular Potts*: the real `cellsInSilico` (NASTJA) simulator; the setup and the screening that fixed it are in Appendix E. Its per-evaluation cost dominates, which is why it uses the longest wall-clock budget.

Computing environment. All experiments ran on the JUWELS Cluster at the Jülich Supercomputing Centre: dual-socket Intel Xeon Platinum 8168 (Skylake) nodes with 48 physical cores and ≈ 94 GB memory per node, connected by Mellanox EDR InfiniBand. The software stack is ParaStationMPI 5.10 (the 2025 GCC toolchain), Python 3.12.3, NumPy 1.26.4, SciPy 1.13.1, mpi4py 4.0.1, and pyABC 0.12.17, with POT (Flamary et al., 2021) for the sliced Wasserstein distance; the asynchronous sampler runs as a propagator in a fork of Propulate. Unless noted, ranks are placed one per physical core, so a “ W -worker” point occupies

W physical cores packed onto $\lceil W/48 \rceil$ fully-reserved nodes. The production experiment of the three cheap benchmarks (Gaussian mean, g-and-k, Lotka–Volterra) instead placed its 48 ranks as 12 per node on four nodes under whole-node reservation, after the single-node submissions failed, which is why its asynchronous throughput is below the packed single-node scaling run’s on the same benchmark (§6.3). The Cellular Potts production run and the straggler, heterogeneity, twin and scaling experiments are single-node or fully packed. Each strong-scaling point reports five replicates (three to five for the largest Cellular Potts points), and figures show medians with inter-quartile ranges.

E The Cellular Potts inference setup

The benchmark runs the `cellsInSilico` Cellular Potts tissue model (Berghoff et al., 2020) through the NASTJA framework (Berghoff et al., 2018): a 50^3 lattice for 501 steps (an 80^3 lattice for 1001 steps in the larger configuration), seeded with a small cluster of cells that divide and move, summarized at the final step. The inference problem was designed on a screening experiment of about 12,000 simulations over space-filling designs of six candidate parameters, scored by the ratio of between-parameter to within-parameter variation of each summary block. Of the seven summary blocks the shipped configuration weighted equally, two scalars, the log cell count and the log 95th-percentile radius, carry nearly all of the signal (signal-to-noise 12 and 9 against about 1 for the four ten-dimensional curve blocks), so the discrepancy is taken over those two, equally weighted. One evaluation is the feature-average of four replicate simulations at the same parameters, because a single realization is too noisy for the discrepancy to order nearby parameters. The two summaries resolve exactly two directions (how many cells there are, and how large the cluster is at a given count), so the inferred parameters are the two that move them independently: the division rate (count) and the target cell volume (radius at fixed count), whose responses are near-orthogonal ($|\cos| = 0.07$). Every other candidate expresses itself through the count: cell motility, adhesion and surface tension are confounded with the division rate at $|\cos| \geq 0.8$ at 50^3 , and at 80^3 , with $6.4\times$ the cells, the confounding sharpens rather than breaks ($0.81 \rightarrow 0.98$ for the surface term). The benchmark is therefore two-dimensional as a property of the model, not for want of screening. Priors are log-uniform on $[0.001, 0.2]$ and $[200, 1200]$ with truths 0.009 and 500; motility is fixed at 1400. Rejection ABC on 2000 prior draws forecast 91% and 64% contraction for the two parameters before any adaptive run, against the 92% and 81% the asynchronous estimator reports (§6.4).

Two configurations. The *production* configuration is the two-parameter setup above and supplies every posterior result, Tables 2 and 4, the 80^3 systems comparison, and the theory diagnostics of Appendix A. The *earlier* configuration was used by the 50^3 twin and scaling experiments (the 50^3 points of Fig. 2c, Table 11, Fig. 3 right, Fig. 13). It inferred the cell division rate and motility on normalized $[0, 1]$ priors mapping to $[6\times 10^{-5}, 0.6]$ and $[0, 10^4]$ (true normalized values 0.0499 and 0.2), scored one simulation per evaluation (5.1 s at 50^3 , in-run coefficient of variation 0.10) by a whitened distance over morphological summary features (radial density and pair-correlation profiles, free-area and structure-factor profiles, a DBSCAN dispersion fraction, and log cell count and radius). The screening above found those two parameters confounded, which is why the production configuration replaced them. The systems measurements depend on the configuration only through its runtime distribution, which the predictor takes from the run itself.

F Additional results

The studies below support the main text and are reported in full here.

F.1 Runtime heterogeneity: archive quality

The synthetic instance. The heterogeneity study shows the same conversion where the runtime law is known (Fig. 7): at a fixed 60 s budget on the Gaussian benchmark pyABC completes fewer simulations as σ grows and its archive’s posterior-mean error to the analytic posterior climbs from ≈ 0.05 to ≈ 0.28 , while the asynchronous sampler completes several-fold more and its archive holds ≈ 0.007 – 0.011 . These are archive means; the posterior estimates of both samplers in this experiment are scored in Table 10.

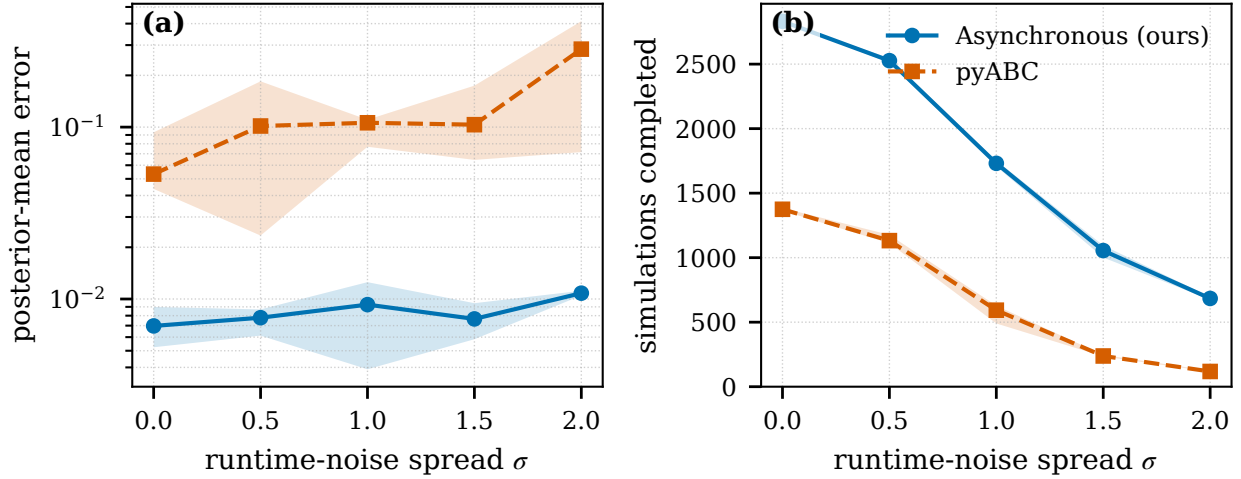


Figure 7: Inference per unit wall clock under runtime heterogeneity (Gaussian mean, analytic posterior, 48 workers, 60s budget, five replicates; bands are inter-quartile ranges). Left: error of the archive mean (the unweighted top- k archive, not $\hat{\pi}_n$; §6.4) to the analytic posterior mean, against the runtime spread σ . Right: simulations completed within the budget. The barrier starves pyABC as σ grows; the asynchronous sampler’s error stays flat.

F.2 The barrierized twin on the straggler benchmark, and pyABC under injected heterogeneity

A persistent straggler. One of 16 workers is given a permanent post-evaluation delay of 0.1s scaled by 0, 1, 5, 10, 20 (Table 9). The slow worker’s recorded evaluation time (2.000s at 20 $\times$) plus the 3.6ms per-evaluation overhead measured on the fast workers is the generation time, so the twin cannot exceed $16/2.004 = 7.98$ simulations per second, and the measured asynchronous rate over that gives 401 $\times$ against a measured 402 $\times$; at 5 $\times$ and 10 $\times$ the agreement is the same. At 0 $\times$ and 1 $\times$ the prediction fails: with no straggler to wait for, the collective itself costs about 0.3s per barrier on a 4ms workload, a term the model does not contain, and coarsening the barrier to one per 112 evaluations (pyABC’s population) amortizes it. From 5 $\times$ up the two barrier granularities agree with each other and with the prediction within 0.5% (0.2% at 10 $\times$, 0.06% at 20 $\times$): once a straggler binds, the barrier’s cost does not depend on how often it is taken.

The schedule must move. The twin shows the bandwidth transient of §6.4 on a cheap simulator. At the twin’s own simulation count and seeds the asynchronous sampler reports $W_1 = 0.010$ and 0.068 to the analytic posterior at 10 $\times$ and 20 $\times$ where the twin reports 0.79 and 1.71 (Table 9, lower block). The propagator is the same, and so are the 200 calls per rank at 20 $\times$. But the twin’s schedule reached a bandwidth of 2.9, against 0.16, on histories whose 100th-smallest discrepancy is 0.008, because under lockstep it takes fewer search steps (Appendix C). Re-reporting the twin’s histories at $\epsilon_{(100)}$ brings them to 0.010–0.015.

F.3 The barrierized twin under runtime heterogeneity and on Cellular Potts

Table 10 gives the twin’s heterogeneity experiment behind §6.1, with the posteriors of every sampler at the bandwidth its schedule reached and re-reported at the order statistic (§6.4); Table 11 factorizes the twin’s Cellular Potts slowdown into the barrier’s share and the unattributed duration ratio.

F.4 Tolerance curves

Figure 9 shows the curves Table 2 is read from.

Table 9: Barrierized twin on the straggler benchmark ($W = 16$, five replicates, medians). Throughput in simulations per second; *predicted* as in §5.3, and at $0\times$ and $1\times$ the collective’s latency, absent from the model, dominates. Lower block: W_1 of each sampler’s posterior to the analytic posterior at the twin’s simulation count and seeds, at the bandwidth its schedule reached and, last row, re-reported at $\epsilon_{(100)}$ (§6.4).

Straggler slowdown	$0\times$	$1\times$	$5\times$	$10\times$	$20\times$
Asynchronous, sims/s	3888	3112	3241	3218	3205
Twin, barrier every W	48.7	65.3	31.6	15.9	8.0
Twin, barrier every 112	162.8	148.4	31.8	15.9	8.0
Twin, predicted	(3870)	(154)	31.8	15.9	7.98
Ratio, measured (every W)	$80\times$	$48\times$	$103\times$	$202\times$	$402\times$
Ratio, predicted	—	—	$102\times$	$202\times$	$401\times$
<i>Posterior recovery, W_1 to the analytic posterior, at the twin’s simulation count</i>					
Evaluations (= calls per rank $\times 16$)	75,104	44,512	12,304	6,304	3,200
Bandwidth reached, asynchronous	5×10^{-4}	7×10^{-4}	0.003	0.006	0.16
Bandwidth reached, twin every W	5×10^{-4}	8×10^{-4}	0.06	1.3	2.9
$\epsilon_{(100)}$ of the same history	3×10^{-4}	5×10^{-4}	0.002	0.003	0.008
Asynchronous at that count	0.011	0.009	0.010	0.010	0.068
Twin, barrier every W	0.013	0.007	0.016	0.789	1.713
Twin, barrier every 112	0.009	0.009	0.009	0.017	1.082
Twin every W , re-reported at $\epsilon_{(100)}$	0.015	0.010	0.010	0.010	0.010

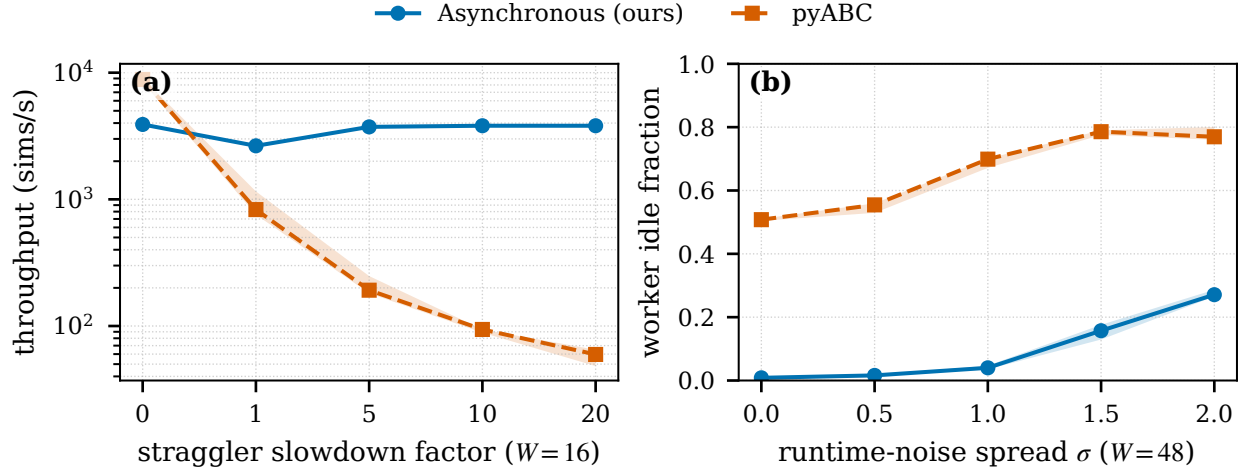


Figure 8: pyABC against the asynchronous sampler under injected runtime heterogeneity on the Gaussian mean (§6.1). (a) Throughput under a persistent straggler: one of 16 workers carries a permanent post-evaluation delay of 0.1s scaled by the factor (300s; medians and inter-quartile ranges over five replicates; logarithmic axis). (b) Fraction of worker time spent idle under a lognormal runtime multiplier of spread σ (48 workers, 60s).

F.5 Parameter-coupled runtime

The heterogeneity study above slows simulations independently of their parameters; a sharper test couples runtime to the *inferred parameter*, probing the completion-time selection of §8. On the Gaussian-mean benchmark (48 workers, 60s budget, five replicates) the post-evaluation delay is $d(\mu) = \min(d_0 e^{c(\mu - \mu_c)/\ell}, d_{\max})$ with $d_0 = 0.02$ s, $\mu_c = 0$, $\ell = 0.15$ and $d_{\max} = 0.5$ s, swept over $c \in \{0, 0.5, 1, 2\}$ ($c = 0$ is uniform runtime), so one region of parameter space evaluates far faster than its mirror image and the every-particle rule over-

Table 10: Barrierized twin under runtime heterogeneity ($W = 48$, lognormal runtime multiplier of scale σ , five replicates, medians of simulations per second of each sampler’s own elapsed wall clock). *Predicted* is from the injected law and the asynchronous run’s measured overhead. The $\sigma = 0$ control costs almost nothing because these evaluations take 1 s and barrier latency is negligible against them; the cost is the wait for the slowest draw and grows with σ . pyABC degrades less than the twin because it discards latecomers instead of waiting for them. The lower block scores the posteriors against the analytic posterior at the bandwidth each schedule reached, then re-reported at the 100th order statistic of the same history (§6.4); no rank made k calls in the asynchronous run, so its schedule never ran at $\sigma \leq 0.5$, and the twin’s ran only at $\sigma \leq 0.5$.

Heterogeneity σ	0	0.5	1.0	1.5	2.0
Asynchronous, sims/s	47.6	42.1	28.9	17.6	11.4
Twin, barrier every W	44.8	15.1	4.37	1.06	0.37
Twin, barrier every 96	45.1	19.8	6.50	1.75	0.45
pyABC	23.1	18.9	9.91	4.35	2.39
Ratio, measured (every W)	1.06×	2.8×	6.6×	16.6×	29.6×
Ratio, predicted	1.00×	2.77×	6.34×	14.0×	35.4×
Ratio, pyABC	2.1×	2.2×	2.9×	4.1×	4.8×
<i>Posterior recovery, W_1 to the analytic posterior (calls per rank: async 59–13, twin 440–10)</i>					
Evaluations, async	2832	2527	1731	1055	684
Evaluations, twin	21,120	6,912	2,112	672	480
Bandwidth reached, async	5	5	0.92	1.25	0.78
Bandwidth reached, twin every W	0.36	2.4	5	5	5
Async, at the schedule’s bandwidth	2.12	2.04	0.68	0.93	0.56
Twin every W , at the schedule’s bandwidth	0.19	1.43	2.20	2.22	2.28
pyABC, last generation	0.53	0.61	1.36	2.19	2.21
Async, re-reported at $\epsilon_{(100)}$	0.011	0.011	0.010	0.013	0.042
Twin every W , re-reported at $\epsilon_{(100)}$	0.015	0.013	0.012	0.040	0.203
Async archive (top 100)	0.018	0.017	0.016	0.014	0.019

Table 11: The twin’s slowdown on Cellular Potts 50³ in the benchmark’s earlier single-simulation configuration (Appendix E), factorized (five replicates, medians). Throughput is utilization over mean simulation time, so the ratio is the product of the two columns that follow. The utilization ratio is what the barrier costs; it agrees with the straggler factor measured inside the twin’s own generations and with the prediction from the asynchronous run’s runtime distribution. The duration ratio (the twin’s simulations running longer, with replicate means from 6 to 36 s against 5.1 s on every asynchronous replicate) is not a barrier effect and is not attributed here.

Workers (= cores)	48	96	192	384
Throughput ratio, async / twin	1.91×	1.87×	1.86×	2.82×
= utilization ratio	1.18×	1.19×	1.21×	1.21×
× simulation-duration ratio	1.63×	1.59×	1.52×	2.33×
Straggler factor of the twin’s generations	1.21	1.28	1.21	1.29
Predicted from asynchronous timing	1.26×	1.30×	1.32×	(2.34×)

represents it in the raw archive. The coupling has the intended effect on the sampling rate: as it steepens, aggregate throughput falls by $\approx 43\%$ for the asynchronous sampler and more steeply for pyABC (Fig. 10a).

The posterior does not follow. The asynchronous posterior-mean error to the analytic posterior stays ≈ 0.01 at every coupling, comparable to pyABC, which discards latecomers (Fig. 10b), with or without AMIS ($S=0$), for the unweighted archive and for the AMIS-reweighted estimator (6), and in a steeper regime (base delay 0.5 s, cap 8 s). The reweighting changes the spread, not the mean: it lowers the effective-sample-size fraction from ≈ 0.99 to ≈ 0.88 and widens the central 90% interval by $\approx 15\%$. A drain-after-deadline variant, which lets every in-flight simulation finish past the deadline, moves the weighted posterior mean by at most

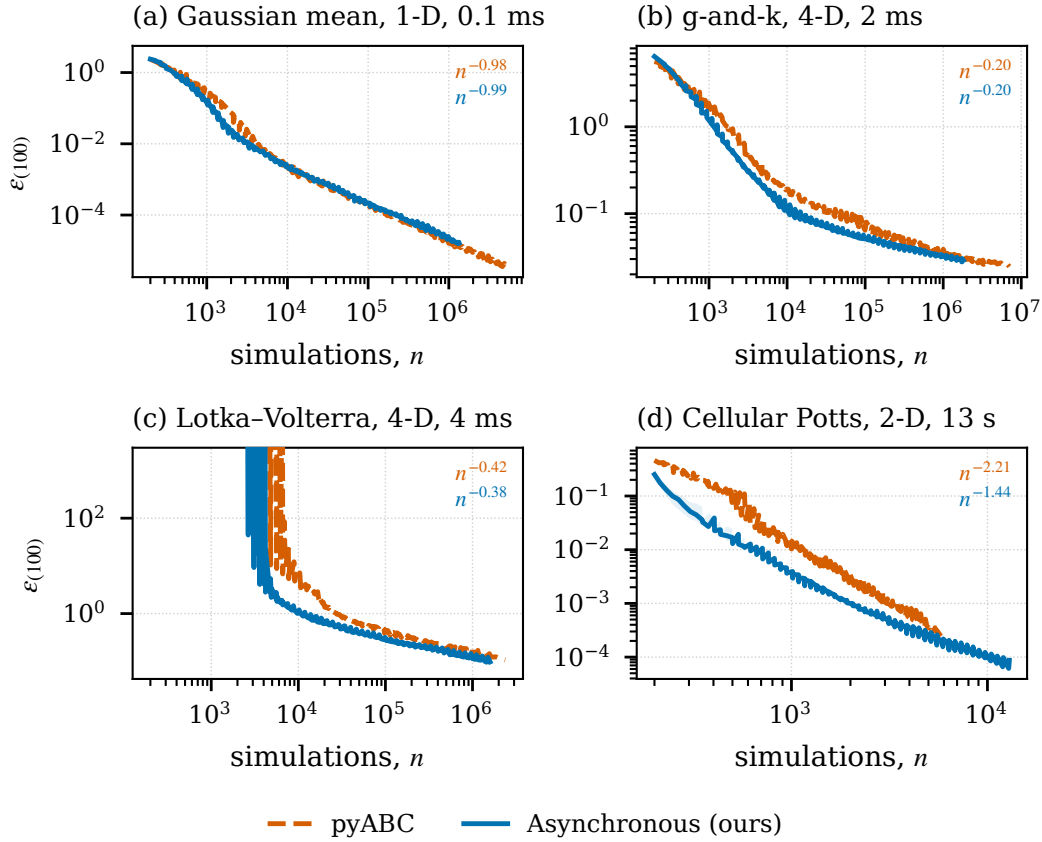


Figure 9: Tolerance against simulations spent, $\epsilon_{(100)}(n)$, asynchronous against pyABC on every benchmark (medians and replicate range over five replicates; the slope over the last decade of n in the legend). The vertical gap at equal n is the per-simulation efficiency, the horizontal extent is the extra simulations the wall clock allowed. On Lotka–Volterra 98% of simulations return the extinction fallback, so $\epsilon_{(100)}$ sits on that plateau until about 5000 draws have produced 100 survivors; the plateau is cut off.

0.013, within the replicate spread and with no systematic direction, so the deadline censoring is negligible at these couplings. The raw archive shows no measurable mean bias at these couplings; AMIS neither creates nor removes one here, and a correction in the manner of anytime Monte Carlo (Murray et al., 2021) is future work.

F.6 Strong scaling on a cheap simulator

Lotka–Volterra is an adversarial simulator for a scaling study. Each evaluation takes ≈ 4 ms, so there is little per-evaluation compute to distribute, and an instrumented run (Fig. 11) shows a worker inside the simulator $\approx 25\%$ of its time at the 48-worker throughput peak, falling to $\approx 8\%$ and $\approx 2\%$ as the job spreads from one node to three to six. The mechanism is the method’s single shared population: every completed evaluation is exchanged with all workers so that each can propose from the current archive, an all-to-all exchange whose per-arrival cost grows with the worker count. Aggregate throughput therefore rises almost linearly while the job fits on one node and then declines across fully packed multi-node jobs (Table 12), because adding ranks beyond one node lengthens the exchange faster than it adds useful work. pyABC is given a population equal to the worker count at high core counts (a smaller population cannot occupy every core); it scales steadily and overtakes the asynchronous sampler beyond three nodes. This is the boundary of the asynchronous advantage on a cheap, homogeneous simulator; it disappears once an evaluation has real cost (Fig. 3, right). The archive size moves the boundary, because the per-arrival proposal reconstruction is $\mathcal{O}(k)$. Raising k to

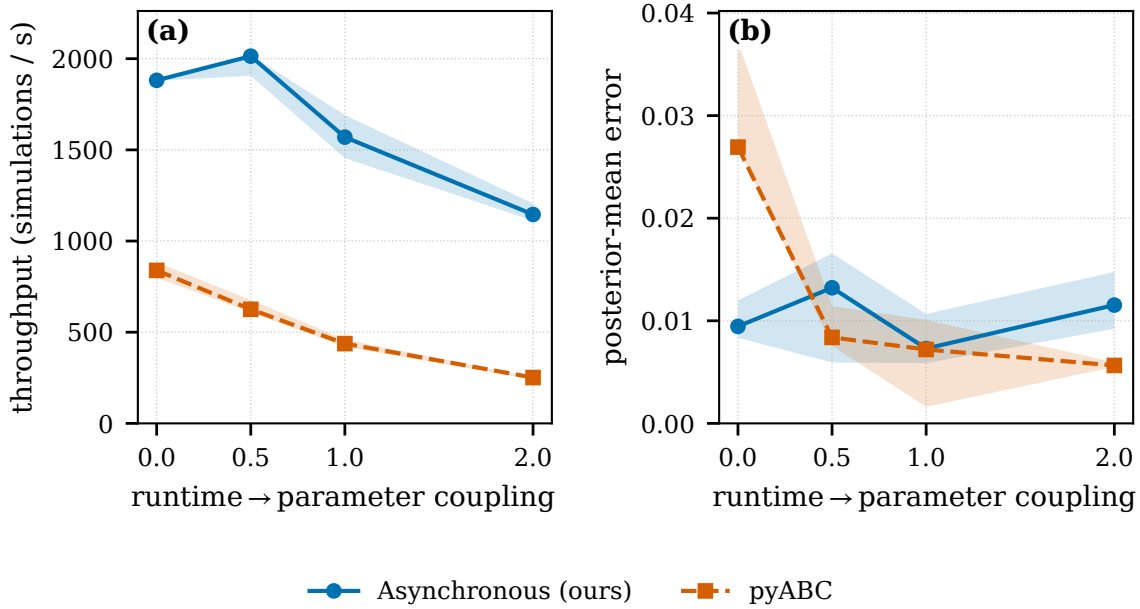


Figure 10: Parameter-coupled runtime on the Gaussian-mean benchmark (analytic posterior; 48 workers, 60 s budget, five replicates; bands are inter-quartile ranges). The post-evaluation delay grows with the inferred mean, so one region of parameter space is sampled far more often than its mirror image. (a) Aggregate throughput falls as the coupling steepens (monotonically for pyABC, and from a peak at $c = 0.5$ for the asynchronous sampler), consistent with the fast region dominating the raw sample counts. (b) Posterior-mean error to the analytic posterior nonetheless does not increase with coupling and stays comparable between the asynchronous sampler and pyABC, which discards latecomers, so the over-representation does not measurably distort the posterior estimate at these coupling levels.

Table 12: Strong scaling on Lotka–Volterra: median simulation throughput (sims/s) at a 180 s budget, over five replicates (Fig. 3, left, shows the inter-quartile spread). Asynchronous (ours) at both archive sizes vs. pyABC. Multi-node points use *fully packed* nodes (worker counts that are exact multiples of 48); at counts above 100 pyABC is given *population size equal to the worker count* so it can occupy every core (a population of 100 leaves the surplus idle), which is why it keeps scaling here. The asynchronous sampler peaks on one node and then declines as multi-node coordination dominates, so pyABC overtakes it beyond three nodes at $k = 100$. Enlarging the archive tenfold costs throughput at every scale, because the per-arrival proposal reconstruction is $\mathcal{O}(k)$, and moves that crossover in from beyond three nodes to three.

Workers	1	4	16	48	144	192	240	288
Nodes	<1	<1	<1	1	3	4	5	6
Synchronous	105	96	437	2952	3757	4322	4539	6034
Asynchronous ($k = 100$)	149	369	1983	6240	4195	2615	2088	1729
Asynchronous ($k = 1000$)	106	133	620	3804	2804	1957	1587	1336

1000 costs 39% of the single-node peak throughput. Sweeping k over $\{50, 100, 200, 400, 800, 1000\}$ resolves that cost into a smooth monotone decline, with a single step in the crossover: from beyond three nodes to three at $k \geq 400$ (Table 14).

F.7 Calibration against the archive size and the snapshot buffer

The calibration tests of §6.4 fix k and S per benchmark, which leaves open whether the two efficiency knobs must be retuned as the dimension grows. To test this we sweep them directly on a d -dimensional Gaussian-

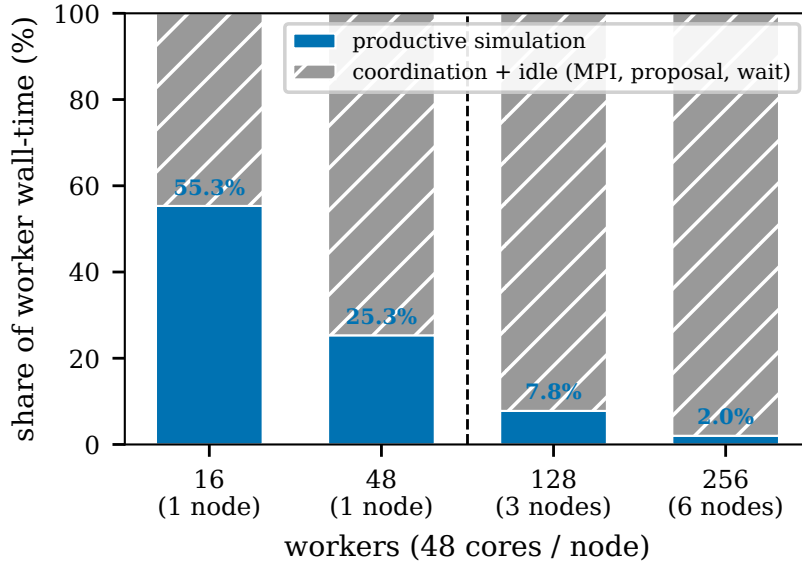


Figure 11: Measured decomposition of asynchronous worker wall-time on Lotka–Volterra ($k = 100$; 180 s budget) from a dedicated instrumented run. The productive simulation fraction (blue, annotated), the share of worker wall-clock spent inside the simulator, collapses from 55% at 16 workers to 2% at 256, while coordination and idle (MPI exchange, per-arrival proposal reconstruction, serialization, waiting) dominate as the job spreads beyond one node (16/48 = 1 node; 128 = 3 nodes; 256 = 6 nodes). The strong-scaling decline of Fig. 3 is therefore a measured property of coordinating a near-instantaneous simulator.

mean target, $y_i \sim \mathcal{N}(\theta, \sigma^2 I_d)$ with $\theta \in \mathbb{R}^d$ under a flat box prior, whose posterior is known exactly at every d (SBC against that exact posterior returns 0.496/0.794/0.905/0.952, so the target itself is calibrated and any departure below is a property of the method). The grid is $d \in \{1, 2, 4, 8, 16\} \times k \in \{50, 100, 200, 400, 800\} \times S \in \{0, 5, 20, 50\}$, 1000 trials per cell, reporting the same full-history estimator (Table 13). S sets both the online buffer length and the number of reconstructed proposals in the posterior denominator (§3.5), and the sweep varies the two together; at $S = 0$ the denominator holds only the first and the last reconstructed proposal, so the $S = 0$ column tests the posterior-side mixture as much as the online buffer. At $S = 0$ the estimator is over-confident in one and two dimensions (mean deviation -0.088); every $S \geq 5$ removes that deviation, and the experiments use $S = 20$.

The archive size does not, in this range, trade calibration against throughput. Because k has both a statistical effect (above) and a systems cost (per-arrival proposal reconstruction is $\mathcal{O}(k)$, so a larger archive is slower at every worker count; §6.3), one might expect to trade throughput for calibration. Over the range we can test, that expectation does not hold. Table 14 puts the throughput cost of k next to its worst-case miscalibration over the dimensions swept, using a throughput grid matched on everything but k (Lotka–Volterra, 180 s wall cap, five replicates, worker counts 1–288: $k \in \{100, 1000\}$ from the main scaling run and $k \in \{50, 200, 400, 800\}$ from a dedicated k -frontier run sharing that configuration). Of the six archive sizes, only two are Pareto-optimal, and they are the two smallest: $k = 100$ is simultaneously the best-calibrated setting tested (worst deviation 0.064, at $d = 4$) and within 7% of the fastest (6240 vs 6684 simulations/s at the single-node peak). Every larger archive is dominated by it on *both* axes: $k = 200$ is 5% slower and nearly three times worse calibrated (0.180), $k = 400$ is 14% slower and 0.195, and $k = 800$ costs 31% of throughput and is still slightly worse calibrated than $k = 100$ (0.074). The only faster archive, $k = 50$, is the one that fails calibration (0.250 at $d = 8$). The archive size therefore needs only to clear a threshold: above the small-archive failure region, further enlargement adds no calibration and costs throughput monotonically.

Table 13: Calibration sensitivity to the archive size k and the AMIS snapshot buffer S on the d -dimensional Gaussian-mean target (1000 trials per cell; entries are the mean signed deviation of empirical from nominal coverage over the 0.50/0.80/0.90/0.95 levels and the d parameters). Negative is over-confident, positive is conservative; 0.000 is exactly calibrated. **Left:** varying k at $S = 20$. **Right:** varying S at $k = 100$.

archive size k (at $S = 20$)					
	50	100	200	400	800
$d = 1$	-0.014	-0.003	+0.006	+0.006	+0.074
$d = 2$	-0.020	-0.006	+0.009	+0.030	+0.072
$d = 4$	+0.025	+0.064	+0.094	+0.133	+0.063
$d = 8$	-0.250	+0.055	+0.180	+0.195	+0.061
$d = 16$	-0.047	+0.028	+0.049	+0.053	+0.055

snapshot buffer S (at $k = 100$)				
	0	5	20	50
$d = 1$	-0.088	-0.001	-0.003	-0.005
$d = 2$	-0.088	-0.001	-0.006	-0.003
$d = 4$	+0.039	+0.068	+0.064	+0.064
$d = 8$	+0.065	+0.006	+0.055	+0.074
$d = 16$	-0.010	+0.010	+0.028	+0.034

Table 14: The archive size on both of its axes at once. **Throughput:** median asynchronous throughput at the single-node peak (48 workers) on Lotka–Volterra, 180s wall cap, five replicates ($k \in \{100, 1000\}$ from the main scaling run, $k \in \{50, 200, 400, 800\}$ from the matched k -frontier run), and its change relative to the $k=100$ we use throughout. **Calibration:** worst absolute deviation of empirical from nominal coverage over the dimensions of the d -dimensional Gaussian-mean sweep at $S=20$ (Table 13), with the dimension attaining it. **Crossover:** fully packed nodes at which pyABC (population = worker count) overtakes the asynchronous sampler. The sweep did not include $k=1000$.

k	sims/s at $W=48$	vs. $k=100$	worst $ \Delta\text{cov} $	at	pyABC overtakes
50	6684	+7%	0.250	$d=8$	beyond 3 nodes
100	6240	—	0.064	$d=4$	beyond 3 nodes
200	5904	−5%	0.180	$d=8$	beyond 3 nodes
400	5393	−14%	0.195	$d=8$	3 nodes
800	4331	−31%	0.074	$d=1$	3 nodes
1000	3804	−39%	—	—	3 nodes

The two axes come from different benchmarks by design (calibration needs a target whose posterior is known at every d ; throughput needs the regime where an $\mathcal{O}(k)$ per-arrival cost is most exposed), so the throughput column is an upper bound on the relative cost of k and the conclusion is conservative. At scale k acts as a step in the crossover with pyABC at a population equal to the worker count: beyond three fully packed nodes for $k \leq 200$, three nodes for $k \geq 400$ (last column).

F.8 The point-mass concentration diagnostic

A Wasserstein distance from the unweighted top- k archive to a point mass at the true parameter is floored at the posterior’s own spread (on the Gaussian configuration an exactly correct posterior scores $0.1\sqrt{2/\pi} = 0.080$), and it ranks a collapsed archive above a correct one. We use it only as a concentration diagnostic, for the ablation and sensitivity grids below, and report none of its values as posterior quality.

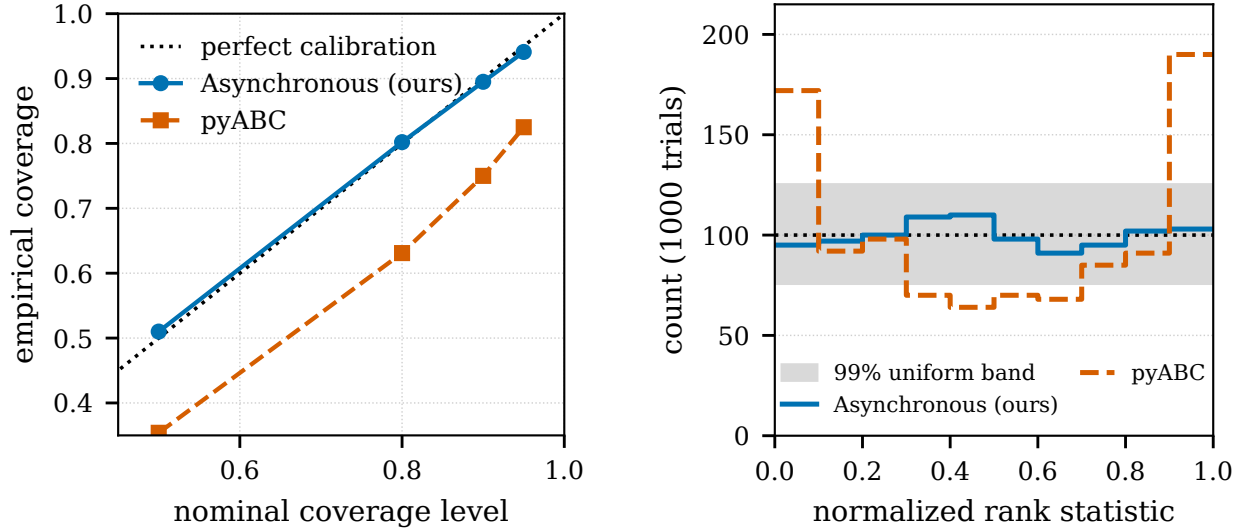


Figure 12: Simulation-based calibration on Gaussian-mean inference (1000 trials; the full-history AMIS posterior): empirical coverage versus nominal level (left) and rank histogram (right). The asynchronous estimator is close to calibrated.

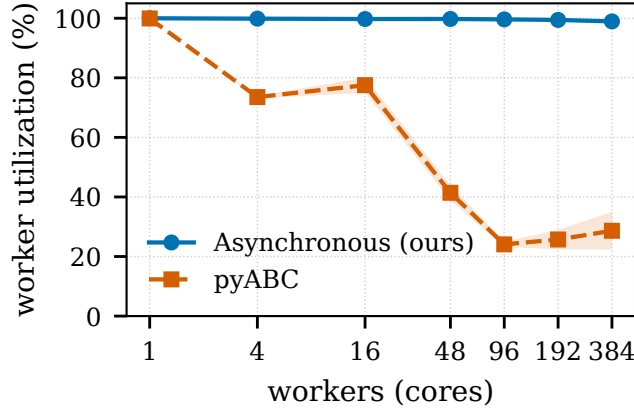


Figure 13: Cellular Potts 50^3 worker utilization in the benchmark’s earlier single-simulation configuration (fraction of worker wall-clock spent inside the simulator) versus worker count (1800s budget; error bars are ± 1 s.d.); pyABC uses a population equal to the worker count (Fig. 3, right). The asynchronous sampler stays ≈ 98.9 – 99.8% utilized at every scale. pyABC fills every core within a generation but then idles at the barrier waiting for the slowest simulation, so its utilization falls to ≈ 24 – 29% ($41\% \rightarrow 24\% \rightarrow 26\% \rightarrow 29\%$ at 48/96/192/384 workers, i.e. ≈ 70 – 76% idle at every multi-node scale). The figure is the measured utilization behind pyABC’s sub-linear scaling in Fig. 3 (right).

F.9 Worker utilization on Cellular Potts at 50^3

F.10 Ablation and sensitivity

An ablation (hard kernel; no AMIS) and a hyperparameter grid (perturbation scale, initial-tolerance multiplier, archive size) on the Gaussian-mean benchmark (48 workers, 300s, five replicates), scored by the concentration diagnostic above, find no ingredient removal or grid setting that changes the archive’s concentration (0.07–0.15 throughout, degrading only mildly toward the largest perturbation scales and initial tolerances). The tolerance scheduler is inert under a smooth kernel, since the bandwidth is chosen by ESS-retention

bisection regardless of the rule, and was held fixed. Removing AMIS is neutral on this uniform-runtime target, consistent with the parameter-coupled study above. Calibration, which this diagnostic cannot see, is the archive-size sweep above.

F.11 Lotka–Volterra extinction

The simulator has an absorbing state: if either population reaches zero before $T_{\max}$ the run returns a fallback discrepancy, which a smooth kernel weights to effectively zero, and at the configured parameters that is the usual outcome (survival 2.6% of 2000 simulations at the true parameter, 1.7% over 400 uniform prior draws). The observed dataset is itself re-drawn until a trajectory survives, so the target is $p(\theta \mid s_{\text{obs}}, \text{survived})$, which is internally consistent: the extinct particles’ zero weight is the $P(\text{survive} \mid \theta)$ factor of the likelihood. Throughput and scaling are unaffected, because an extinct simulation costs what a surviving one costs. Posterior recovery is affected: the informative sample is $\approx 2\%$ of the nominal budget, so we report Lotka–Volterra as a systems benchmark and treat its posterior curves as diagnostics that bound no method’s attainable accuracy.